

Journal of Statistical Software

MMMMMM YYYY, Volume VV, Issue II. *doi: 10.18637/jss.v000.i00*

Imputation of Incomplete Multilevel Data with R

Hanne I. Oberman*
Utrecht University

Johanna Muñoz*
University Medical Center Utrecht

Gerko Vink
Utrecht University

Thomas P.A. Debray
Smart Data Analysis and Statistics B.V

Valentijn M.T. de Jong
University Medical Center Utrecht

Abstract

The analysis of multilevel data requires careful attention when some measures are missing. A common strategy is to use imputation and replace missing values with one or more plausible values. The resulting datasets can then be analyzed using standard software packages. Although numerous methods for multiple imputation have been proposed, software implementations for multilevel data are still relatively scarce.

In this tutorial, we introduce the key challenges of dealing with (incomplete) multilevel data and illustrate how to impute them using the R package **mice**. We explain how clustering can be effectively addressed and describe methods for handling non-ignorable missingness mechanisms. To make it easier to use the proposed methods, we provide code snippets and relevant example data for three case studies.

First, we estimate the association between the pupils' sex and their popularity score in data on 2000 pupils' popularity in 100 schools. Second, we estimate the prognostic value of Computerized Tomography on 6-month mortality in data on 11,002 patients across 15 studies on traumatic brain injuries with data Missing At Random. Third, we estimate the association between family obesity and BMI in data on 2,111 patients from 5 regions with data Missing Not At Random.

Keywords: missing data, multilevel, clustering, **mice**, R.

1. Introduction

Missing data frequently occur in observational (non-interventional) studies, and to a lesser extent in experimental studies, like Randomized Controlled Trials (RCTs). Dealing with incomplete data sets, requires careful consideration, not only of assumptions about the cause of the missing data, but also of those intrinsic to the main analysis of the study. This is necessary to prevent bias and reduced statistical power, which could ultimately lead to erroneous conclusions.

Many studies are based on inherently clustered data. For example, RCTs and observational studies often involve patients sampled from multiple sites, such as countries or hospitals. Electronic Health Records datasets frequently collect information from multiple hospitals, and some studies even rely on aggregate data from multiple studies, such as Individual Participant Data (IPD).

Commonly, these studies face two simultaneous challenges: 1) the cluster structure and 2) missing data. The observations within clusters are not independent, which requires the use of multilevel modeling techniques in the main analysis to account for the hierarchical structure within the data. This structure must also be considered when dealing with missing data, as several imputation methods are based on models with implicit assumptions. Therefore, we will briefly explain the concepts of both cluster structure and missing data, and discuss why their presence can be problematic.

This manuscript serves as a tutorial for imputing incomplete multilevel data using the **mice** (Multivariate Imputation by Chained Equations) package in R. **mice** has emerged as the de facto standard for imputation via chained equations, which addresses missingness iteratively on a variable-by-variable basis. It is recognized for producing valid inferences across a range of missing data scenarios (Van Buuren 2018).

We provide a practical guide and code snippets for various missing data scenarios, encompassing non-ignorable mechanisms. Through three case studies from the fields of social and biomedical sciences, we demonstrate the practical use of software tools for this purpose. For brevity, our focus lies exclusively on multilevel imputation via chained equations with **mice**; other imputation methods and packages (e.g., **jomo** Grund, Lüdtke, and Robitzsch (2018), **smcfcs** Bartlett, Keogh, and Bonneville (2022), **mdmb** Grund, Lüdtke, and Robitzsch (2021)) fall beyond the scope of this tutorial. We only consider multilevel data with two levels, and we also assume basic familiarity with the notation for the multilevel models in the R packages **lme4** Bates, Maechler, Bolker [aut, cre, Walker, Christensen, Singmann, Dai, Scheipl, Grothendieck, Green, Fox, Bauer, and copyright on simulate.formula) (2022) and **nlme** Pinheiro, Bates, DebRoy, Sarkar, and Heisterkamp (2023).

1.1. Motivating datasets

We illustrate imputation of incomplete multilevel data using three case studies:

- Individual participant data of patients suffering from traumatic brain injury (`impact`) retrieved from the **metamisc** package Debray and de Jong (2021) (empirical data on traumatic brain injuries, $n = 11,022$ patients across $N = 15$ studies with data that are missing at random, MAR);

*Contributed equally

- Hox pupil popularity data with missing popularity scores (`popmis`) from the **mice** package (van Buuren and Groothuis-Oudshoorn 2021) (simulated data on perceived popularity, $n = 2,000$ pupils across $N = 100$ schools with data that are MAR);
- Data from an obesity survey across different sites (`obesity`) retrieved from the **micemd** package Audigier and Resche-Rigon (2023) (simulated data on obesity, $n = 2,111$ patients across $N = 5$ regions with data that are missing not at random, MNAR).

For each of these datasets, we guide the user through a simplified imputation workflow, highlighting the technical aspects of a chosen imputation strategy. The selection of the strategy is based on the nature of the missing data,

1.2. Concepts in multilevel data

Many datasets include individuals that are clustered together, for example in geographic regions, or different studies. In the simplest case, individuals (e.g., students) are nested within a single level of cluster (e.g., school classes). More complex clustered structures may occur when there are multiple hierarchical levels (e.g., students in classes in different schools or patients in hospitals in regions across countries), or when the clustering is non-nested (e.g., electronic health record data from diverse settings and populations within large databases).

With clustered data it is often the case that individuals from the same cluster are more similar than individuals from other clusters. In statistical terms, this implies that observations from the same cluster are not independent and may, in fact, be correlated. If this correlation is left unaddressed, estimators for p-values, confidence intervals, and model parameters are prone to bias (Localio, Berlin, Ten Have, and Kimmel 2001). Statistical methods for clustered data typically adopt multilevel models that explicitly describe the grouping of observations. These models are also known as 'multilevel models', 'mixed effect models', 'random effect models', and in the context of time-to-event data as 'frailty models'. Box 1 provides an overview of multilevel data and Box 2 provides an overview of some key concepts in multilevel modeling.

Box 1: Multilevel data

Data are grouped into clusters at different levels, observations belonging to the same cluster are expected to share certain characteristics. We can specify:

- **Sample units:** units of the population from which measurements are taken in a sample, e.g., student or patient.
- **Cluster:** variable that specify the cluster or grouping, e.g., classroom or hospital.
- **Level-1 variable:** variable that varies within a cluster, e.g., test score.
- **Level-2 variable:** variable that does not vary within a cluster but between clusters, e.g., teacher experience or geographical region of the hospital.

Box 2: Multilevel model

A multilevel (hierarchical) model is a statistical model appropriate for situations where observations are organized at more than one level (e.g., groups). This type of model recognizes and explicitly accounts for the potential correlation of observations within these groupings, addressing the dependencies that often violate the independence assumption between observation of traditional regression models. To achieve this, multilevel models may incorporate fixed or random effects:

- **Common effect:** Also called a fixed effect. It is an effect that is assumed to be constant across all sample units, e.g., gender effect that is the same for all classes or hospitals.
- **Random effect:** Effects that are a source of random variation in the data, and whose levels are sampled from a distribution. e.g., the test score may vary between classrooms or hospitals due to uncontrolled factors such as the teacher experience or hospital protocols.

In R, multilevel models can be fitted using packages such as **lme4** (Bates *et al.* 2022) or **nlme** (Pinheiro *et al.* 2023). In addition to random intercepts, multilevel models can also include random slopes and heterogeneous residual variances across clusters (see e.g., (Gelman and Hill 2006; Hox, Moerbeek, and van de Schoot 2017; de Jong, Moons, Eijkemans, Riley, and Debray 2021)).
The variability due to clustering is often measured by the **intraclass correlation (ICC)**, which in simple terms describes how similar the units in the same cluster are.

1.3. Concepts in missing data

In a study, any variable can have missing values, including those in the main model, such as exposures, outcomes, and confounders, as well on other variables, such as **auxiliary** variables, which, although not explicitly in the analysis model, can improve the imputation of missing values.

These incomplete variables can be categorized by a **missingness pattern** that describes how observed and missing values are arranged in the dataset. There are different types of missing patterns such as univariate (single variable with missing data), monotone (for a participant, all values after a certain time are missing), and file matching (one variable is missing whenever another is observed) (Van Buuren 2018; **?**) (§ 4.1). Specifically for clustered data, we highlight the difference between sporadic and systematic patterns. Sporadic missingness arises when variables are missing for some but not all of the units in a cluster (Jolani 2018). For example, it is possible that test results are missing for several students in one or more classes. When all observations are missing within one or more clusters, data are said to be systematically missing. These missingness patterns are visualized in Figure 1, where the column X_1 is completely observed, column X_2 is systematically missing in cluster 2, and column X_3 is sporadically missing.

	cluster	X_1	X_2	X_3	...	X_p
1	1			NA		
2	1					
3	2		NA			
4	2		NA	NA		
5	3					
...						
n	N					

Figure 1: Sporadic and Systematic Missingness

To analyze these incomplete data, we have to take the nature of the missingness and the cluster structure into account. For example, the sporadic missingness in X_3 could be easily amended if this would be a level-2 variable (and thus constant within clusters). We could then extrapolate the true (but missing) value of X_3 for unit 1 from unit 2, and the value for unit 4 from unit 3. If X_3 would instead be a level-1 variable (which may vary within clusters), we could not just recover the unobserved 'truth', but would need to use a stochastic imputation data method, or discard the incomplete units altogether (i.e., complete case analysis, CCA).

CCA may introduce bias when the observed values are not a random sample of the population. For example, in a clinical trial comparing a new treatment with a standard or reference treatment, some participants receiving the new treatment may drop out due to side effects. If the sickest patients (with the worst outcomes) have the most side effects, then the sickest patients are excluded under CCA, resulting in a selection bias in which the effect of the new treatment is overestimated. Moreover, reducing the sample size can impede the estimation procedure of the main model, leading to inefficient estimations and reduced statistical power. This may have an impact on the conclusions drawn from hypothesis tests.

Apart from CCA, there are several strategies that can be used to handle missing data, such as maximum likelihood, Bayesian estimation and imputation. Imputation, in particular, separates the missing data problem from the analysis, allowing the completed data to be analyzed as if it were completely observed. It is generally recommended to impute the missing values more than once (multiple imputation) with random variation in order to preserve uncertainty due to missingness and to allow for valid inferences (Rubin 1976).

Imputing missing data requires consideration of the mechanism behind the missingness. Rubin proposed to distinguish between data that are missing completely at random (MCAR), data that are missing at random (MAR) and data that are missing not at random (MNAR; see Box 3). For each of these three missingness generating mechanisms, different imputation strategies are warranted ((Yucel 2008; Hox, van Buuren, and Jolani 2015)).

Box 3: Missingness Mechanism

- **Missing Completely At Random (MCAR):** the probability of data being missing is equal across all entries in the dataset.
- **Missing At Random (MAR):** the likelihood of data being missing depends on observed information within the dataset.
- **Missing Not At Random (MNAR):** the probability of data being missing depending on unrecorded information, making the missingness non-ignorable (Rubin 1976).

In this manuscript, we focus on the multiple imputation (MI) approach using the fully conditional specification (FCS). This approach is popular, easy-to-use and it can be applied to various types of incomplete variables and missingness patterns. The R package **mice** implements this approach. It assumes MAR by default, but it can also be extended to certain MNAR data.

MI assumes that data follow a multivariate distribution, so incomplete values for each variable are imputed (filled in) with draws from this distribution. The FCS approach relaxes this assumption by specifying the multivariate distribution through conditional distributions. The procedure generates imputed values for each variable iteratively, where an **imputation method** needs to be specified for each incomplete variable, conditional on other variables included in the dataset.

MICE and other **MICE**-based packages offer a variety of parametric methods that implicitly assume a (statistical) **imputation model**, as well as non-parametric methods such as Predictive Mean Matching (PMM), Random Forest, or k-Nearest Neighbors (k-NN) to generate the imputed values. These methods can be used for single-level data (defined as 1L methods) as well multilevel data with two levels (2L methods) or more (Grund *et al.* 2018).

After the imputation procedure (or **imputation step**), multiple complete datasets, say M in total, are obtained, with missing values filled in with plausible values. Then, in an **estimation step**, the main model or estimation procedure of the study is applied to each of the M complete datasets. Finally, in a **pooling step**, the results of the M analyses, usually point estimates and their standard errors, are combined into a single result using Rubin's rules.(Rubin 1987)

2. Multilevel imputation strategies

In multilevel incomplete data, the use of imputation strategies that ignore clustering, i.e., 1L methods, might result in biased parameters and underestimation of the between-cluster variance, leading to inflated Type I error rates, especially when the true intra-class correlation (ICC) is large (ICC $>$ 0.1). (Taljaard, Donner, and Klar 2008; Enders, Mistler, and Keller 2016). They should only be used when the proportion of missingness is low (less than 5%) and the ICC is low (ICC $<$ .1) (Grund *et al.* 2018). Therefore, it is crucial to utilize strategies that account for clustering, such as the use of a cluster indicator, which is employing imputation separated by clusters, or adopting multilevel imputation methods (2l-method) (Eddings 2024).

In the **cluster indicator** strategy, the cluster variable (transformed internally by **mice** to dummy variables) serves as a predictor using 1L imputation methods. This implies that all variability between clusters is attributed to the cluster, rendering the inclusion of level 2 variables unnecessary in the prediction model (McNeish and Kelley 2019). This strategy performs well in datasets with few clusters and numerous observations per cluster (Graham 2009). It can be used for estimation of multilevel model parameters when the primary interest of the study is on the fixed effects. However, since the imputed values per cluster are biased towards the cluster mean, it may lead to overestimation of the between-cluster variance and an inflation of the Type II error, particularly when the true ICC is small (ICC < 0.1), the percentage of missingness is large, and with small cluster sizes (Andridge 2011; Drechsler 2015).

In the **separate cluster imputation** strategy, the data set is divided into subdatasets according to the cluster variable. Then, imputations are separately performed on each subdataset, and combined (e.g., using the `rbind` function) in one imputation object that is finally used in the main analysis. As the imputation models are independently estimated in each cluster, and in some cases may use different predictors, e.g., when a predictor variable is systematically missing as the information was not collected in a group. Similar to the cluster indicator approach, this method could lead to an overestimation of the variance between clusters in the estimation of a multilevel model and, therefore, a reduction in power. %This strategy is sometimes used when the interaction term between variables is of special interest in the main analysis, for example, the interaction between a treatment and a group (Van Buuren 2018).

In the **multilevel imputation** strategy, 2L methods are used that have been developed to internally account for the multilevel structure of the data. This allows for proper estimates of the fixed effects and the variance components of the multilevel model (Speidel, Drechsler, and Sakshaug 2018). The 2l methods include **1-stage methods**, where a multilevel model is estimated using the complete observations. This model associates the target variable (to be imputed) with a set of predictor variables, incorporating both fixed and random effects. While these methods tend to perform better for balanced data, their performance may be compromised when the number of clusters is insufficient to accurately estimate the multilevel model (Grund *et al.* 2018).

Additionally, there are **2-stage methods** where the imputation model is estimated in two steps. In the first step, a model (e.g., a generalized linear model) is estimated for each cluster based on the complete observations, and in the second step, the parameters of the models for each cluster are combined using meta-analysis techniques to obtain a model with pooled parameters, which serves as the basis for imputing missing values. Some of the 2L methods used in `mice` are detailed in the Appendix, Section A1.1.

It has been recommended that the imputation strategy should align with the main (study analysis) model, commonly referred to as congeniality (Meng 1994). Hence, it's vital to carefully evaluate the assumptions of each method and compare them with those of the main model. This includes considerations such as assumptions about the distribution of the outcome model, inclusion of interaction terms, and homoscedasticity (equal error variance across clusters). However, it's essential to examine the information provided by the data and determine if it sufficiently meets the requirements imposed by the imputation method, taking into account factors like sample size, number of clusters, and missing patterns (e.g.,

systematic missingness). Additionally, it's worth considering whether a complex imputation model is suitable for the data, as less complex strategies may also yield unbiased estimates in specific scenarios (Bailey, Andridge, and Shoben 2020).

3. Imputation workflow

Below, an imputation workflow is described that can be generally used to impute multilevel data. A brief description of each step is provided, along with general recommendations about the technical procedure.

3.1. Pre-imputation

First, the variables in the data set should be clearly identified in terms of their multilevel type (level-1, level-2 or cluster) and variable type (continuous, binary, factor, count). This step will help to select the appropriate imputation method for each type of variable, to correctly specify and include the variables in `mice`, and to ensure consistency in the format of the variables across all clusters.

Next, the observed data should be characterized. We can use a density plot for each cluster or spaghetti plots to understand if the intended main analysis is appropriate for the data (e.g., random intercept vs. a random slope model). Additionally, the ICC should be examined to assess the cluster differences and to select the most convenient imputation strategy (such as cluster indicator versus multilevel methods). Then, the missingness should be characterized using visual tools such as **ggmice** (Oberman 2023). Constructing missingness pattern plots and missingness percentage plots per cluster is recommended to identify the information available at each cluster and also to select the most appropriate imputation method.

When preparing the dataset for imputation, it should be ensured that all essential variables required for the main analysis are included, such as the exposure or intervention, outcome, covariates, confounders, stratification factors, the cluster variable, and sample unit ID. Additionally, other useful variables, such as instrumental or auxiliary variables, should be considered for inclusion, which may enhance the accuracy of the imputation model.

Auxiliary variables with a high proportion of missing data may have minimal impact on the imputation model and could mitigate the risk of multicollinearity or computational issues, especially with certain parametric imputation methods. Therefore, the selection of auxiliary variables should be guided by the association between the incomplete variable and predictor variables, using correlation statistics, as well as the potential contribution of one variable to impute another, using missingness pattern statistics such as the influx and outflux criteria. The inflow criteria quantify the connections between missing data in one variable and observed data in others, while the outflow criteria measure the connections between observed values in one variable and missing data in others (Van Buuren 2018) (§ 4.1.3).

Finally, before imputation is undertaken, we should assess if the estimation procedure is robust to missing data, i.e., if the main model can handle missing data. Sometimes, simpler methods like complete case analysis might suffice, especially if the missingness is low (usually <5%). There are scenarios in which specific Maximum Likelihood (ML) estimation methods outperform Multiple Imputation (MI) methods. For instance, when the response variable is

the only incomplete variable, mixed models demonstrate robustness to missing data under the MAR assumption and with a correct variance-covariance specification (Molenberghs and Kenward 2007).

3.2. Imputation

Choice of individual imputation methods

The imputation model for each incomplete variable in the dataset should be chosen first. The mice package provides a range of imputation methods designed specifically for multilevel data (See Appendix, Section A1.1). Notable among these are **jomo** (Quartagno, Grund, and Carpenter 2019), **micemd** (Audigier and Resche-Rigon 2023), and **pan** (Schafer and Yucel 2002). For a comparison of these methods for addressing variables with systematic and sporadic missingness, see (Audigier, White, Jolani, Debray, Quartagno, Carpenter, van Buuren, and Resche-Rigon 2018).

The `make.method` function in **mice** suggests the most plausible 1L imputation method for incomplete variables according to their type (continuous, binary, categorical). However, modifications are required for clustered data and when the type variable is not aligned with the data (e.g., some skewed continuous variables are better imputed with pmm methods).

To address level-2 variables, the **micemd** package offers the `find.defaultMethod` function. This function assists with selecting a 2L imputation method from the **micemd** methods based on cluster size, the proportion of missing data in each cluster, and the variable class. For level-2 variables, the same value needs to be assigned to all units in the cluster. This can be achieved by specific methods where separate imputations are performed for level-1 and level-2 variables. (Grund *et al.* 2018)

Besides the package-defined imputation methods, custom methods can be specified using the "I formula." This allows deterministic variables to be calculated during the imputation or the imputation methods to be tweaked based on specific conditions, such as conditioning the imputation model on the level of a covariate (e.g., a pregnancy test for females). **Deductive imputation** is used to impute values that can be deterministically deducted from other variables using logical rules or formulas, such as BMI from weight and height. This can also be useful for getting values for level-1 variables from level-2 ones, like in IPD from multiple studies. For example, deductive imputation can be employed to infer missing test values for patients associated with a centre unit that lacks a test procedure or machine, considering accessible metadata or protocol details.

Model specification

The imputation model for each incomplete variable must be aligned with the main (substantive) model. Therefore, it should be ensured that the imputation model shares the assumptions with the main model, such as the treatment of clustering, the inclusion of non-linear or interaction terms, and variable transformations.

However, the imputation model can include additional variables without causing compatibility issues. For example, it's recommended to include the outcome variable in the imputation

model for predictor variables (Moons, Donders, Stijnen, and Harrell 2006). If the outcome is time-to-event, the Nelson-Aalen estimate of the time to the event and the event indicator should be added as covariates in the imputation model (White and Royston 2009). Also, including auxiliary variables associated with the probability of missing data may enhance estimation efficiency (Hardt, Herke, and Leonhart 2012).

Box 4: Prediction Matrix

We assume the following multilevel model with an outcome variable y_{ij} for the i-th cluster and j-th observation associated with $X_{ij} = (x1_{ij}, ..., x5_{ij})$:

$$y_{ij} = (\beta_0 + b_{0_i}) + \beta_1 x_{1_{ij}} + \beta_2 x_{2_{i.}} + (\beta_3 + b_{3_i}) x_{3_{ij}} + \beta_4 x_{4_{ij}} + \beta_{m4} \overline{x}_{4_{i.}} + (\beta_5 + b_{5_i}) x_{5_{ij}} + \beta_{m5} \overline{x}_{5_{i.}} + \epsilon_{ij}$$

The prediction matrix to be included in the `mice` function can be defined according to the following codes:

Code	Definition
-2	Cluster variable, in this case the one defined by j index
0	No predictor variable
1	Fixed effect predictor, e.g., level-1 $x_{1_{ij}}$ or level-2 $x_{2_{i.}}$
2	Random effect predictor, e.g., $x_{3_{ij}}$
3	*Fixed effect predictor with cluster mean, e.g., $x_{4_{ij}}$
4	*Random effect predictor with cluster mean, e.g., $x_{5_{ij}}$
Heckman model:** all predictor variables incorporate random effects	
2	Random effect predictor is included in both the selection and outcome equations
-3	Random effect predictor is only in selection model (Exclusion restriction)
-4	Random effect predictor exclusively appears in the main model.

*For 1-stage 2l methods, it has been recommended to use variables of prediction code 3 and 4, as the inclusion of the means of the cluster might be beneficial for imputation performance. (Mistler and Enders 2017),(Van Buuren 2018)(§ 7.5.1).
**For the Heckman imputation model, code 2 is redefined, and two extra codes are added.

Imputation models are specified on a incomplete variable basis, either using a prediction matrix in the `pred` parameter or through a list of formulas in the `formula` parameter. In the prediction matrix option, the prediction code of each predictor variable is specified for each incomplete variable using predictor codes (see Box 4). On the other hand, the formula option is useful in specifying complex imputation models with polynomial terms or interactions and compared with the prediction matrix method do not require the inclusion of additional terms as Just Another Variable (JAV) (Van Buuren 2018) (§ 6.4).

Recipes have been proposed for setting the imputation models of incomplete level-1 and level-2 variables (Van Buuren 2018) (§ 7.10). These guidelines can be convenient to follow when the main model is complex, such as those with interaction terms at different levels.

Following these recipes helps mitigate convergence problems that may arise due to over-specification in the imputation model and ensures compatibility among the conditionally specified imputation models, as well as congeniality between the imputation model and the main model. However, they can not be applicable to all 2l imputation methods. For instance, the `2l.2stage` imputation methods of the **micemd** package only allow the inclusion of random predictor variables (prediction code = 2).

3.3. Issues during imputation

During the imputation process, certain issues may arise that halt the process. In multilevel model imputations, many issues are related to over-fitting the imputation model. To troubleshoot, it is recommended to inspect the imputation log file for variables causing problems. One approach to address this is to reduce the number of predictors by using functions like `quickpred` or by applying the previously referred recipes step by step.

Another option is to consider variable transformations, such as centering within clusters, and scaling when the model is invariant to linear transformations e.g., random intercept models. Also, adjusting the level of the multilevel model (e.g., using a homogeneous variance imputation method) might also be beneficial.

It is crucial to check the range of imputed variables, as excessively large imputed values for one predictor may trigger convergence issues in other variables. To tackle this, including post-processing specifications on problematic variables or utilizing imputation models like Predicted Mean Matching (PMM) can ensure that imputed values align with observable values.

In some situations, it may be worth considering adopting a strategy that applies separate imputations per study objective. For example, in analyses involving multiple endpoints, conducting distinct imputation processes for each endpoint could be more effective than a unified imputation approach.

3.4. Convergence and Sensitivity Analysis

Before starting the analysis of each imputed dataset, it is crucial to validate the convergence of the imputation process. This is commonly accomplished through trace plots that depict the mean and variance of the incomplete variables across iterations. These plots serve to uncover the need for additional iterations and also to identify potential circular dependencies, i.e., when the missing part of a variable X^{miss} depends on the missing part of another variable Z^{miss}, which in turn depends on X^{miss}. Additionally, it is also important to verify that imputed values fall within a plausible range and also to check the distribution of imputed variables, ensuring that the imputed variable distribution aligns with the distribution of observed values (under the MAR assumption). An alternative approach involves assessing the prediction accuracy of the imputation method (Cai, Van Buuren, and Vink 2023).

While the majority of **MICE** methods are based on MAR assumptions, field expert input may suggest that the MNAR mechanism could be plausible for certain variables. An MNAR variable is one in which the probability of missingness depends on an unobservable variable. This can occur when missingness is associated with the incomplete value itself (self-marking) or when there is an unobserved variable linked to both the value and the probability of

missingness of the incomplete value (indirectly non-informative). Specifically, for the indirectly non-informative case in multilevel datasets, imputation methods based on the Heckman method can be considered (Hammon and Zinn 2020; Hammon 2022; Muñoz, Efthimiou, Audigier, De Jong, and Debray 2023a).

3.5. Main analysis

Finally, after verifying the imputation process, the main analysis is performed on each of the M imputation datasets from the mice output. These calculations can be done manually or easily by passing the mice output object directly into the multilevel model functions of the **nlme** or **lme4** packages.

Then, the model parameters of each imputed dataset are pooled following Rubin's rules (Rubin 1987), which have already been implemented in R functions such as `pool` from the **broom.mixed** package (Bolker, Robinson, Menne, Gabry, Buerkner, Hua, Petry, Wiley, Kennedy, SE), Patil, and Arel-Bundock 2021) or `testEstimates` from the **mitml** package (Grund, Robitzsch, and Luedtke 2023).

4. Case studies

In this section, we will go over the different steps involved in imputing incomplete multilevel data using the `mice` package. Initially, we illustrate how to apply multiple imputation strategies on the `impact` data. Then, using the 'popular' data, we demonstrate how to apply the 1-stage 2L imputation methods along with the application of recipes for interaction terms. Finally, with the obesity data, we apply the 2-stage imputation methods and show how to extend these methods in the case of MNAR data.

4.1. Application of multilevel imputation strategies in Impact

The first case study uses the `impact` dataset, which provides information from 15 studies containing the patient mortality after suffering from traumatic brain injury (TBI) and some admission information values. The dataset includes the following 11 variables:

- `id` Patient id,
- `name` Name of the study,
- `type` Type of study (RCT: randomized controlled trial, OBS: observational cohort),
- `age` Age of the patient,
- `motor_score` Glasgow Coma Scale motor score, factor with 4 levels,
- `pupil` Pupillary reactivity, factor with 3 levels,
- `ct` Marshall Computerized Tomography classification, factor with 3 levels,
- `hypox` Hypoxia (0=no, 1=yes),
- `hypots` Hypotension (0=no, 1=yes),
- `tsah` Traumatic subarachnoid hemorrhage (0=no, 1=yes),
- `edh` Epidural hematoma (0=no, 1=yes),
- `mort` 6-month mortality (0=alive, 1=dead).

This data was previously used by (Steyerberg, Mushkudiani, Perel, Butcher, Lu, McHugh, Murray, Marmarou, Roberts, Habbema, and Maas 2008; Steyerberg, Nieboer, Debray, and van Houwelingen 2019), where models of different complexity were estimated to predict 6-month mortality using the admission information of patients. For simplicity, we considered the common effect approach with respect to the predictor effects β, where the intercepts α vary between clusters, but the predictor effects β are constant. The prediction (main) model uses age, motor_score, pupil reactivity, and CT as predictors. The prediction model for the i-th patient from the j-th study is given as:

$$\begin{aligned} mort_{ij} &= Bernoulli(p_{ij}) \\ logit(p_{ij}) &= \alpha_j + \beta_1 age_{ij} + \beta_2 motor_score_{ij} + \beta_3 pupil_{ij} + \beta_4 ct_{ij} \qquad \text{(main model : 1)} \end{aligned}$$

Here, α_j is the main effect of the cluster j and the effects β_1, β_2, β_3 and β_4 are assumed constant across the clusters.

Pre-imputation

The `impact` data, directly retrieved from the **metamisc** package, is a complete data set. For the purpose of this tutorial we have induced missingness (mimicking the missing data in the original data set before imputation). The resulting incomplete data can be accessed from zenodo. We start by loading the incomplete data into the R workspace and checking the structure:

```
dati <- readr::read_rds("../Data/impact_incomplete.rds")
dati$id <- 1:nrow(dati)
str(dati)
```

```
'data.frame':   11022 obs. of  12 variables:
 $ name       : Factor w/ 15 levels "APOE","CSTAT",..: 13 13 13 13 13 13 13 13 13 13 ...
 $ type       : Factor w/ 2 levels "OBS","RCT": 2 2 2 2 2 2 2 2 2 2 ...
 $ age        : int  53 14 30 35 18 20 26 17 75 19 ...
 $ motor_score: Factor w/ 4 levels "1/2","3","4",..: 4 3 4 4 4 4 1 3 4 3 ...
 $ pupil      : Factor w/ 3 levels "Both","None",..: 1 1 1 1 1 1 1 1 1 3 ...
 $ ct         : Factor w/ 3 levels "I/II","III","IV/V": 1 1 1 3 1 2 1 2 1 1 ...
 $ hypox      : int  0 0 0 0 1 0 0 0 0 NA ...
 $ hypots     : int  1 0 0 0 0 0 1 1 0 0 ...
 $ tsah       : int  0 1 1 0 0 0 1 0 0 0 ...
 $ edh        : int  0 0 0 0 0 NA 0 0 0 0 ...
 $ mort       : int  1 0 0 0 1 0 1 1 0 1 ...
 $ id         : int  1 2 3 4 5 6 7 8 9 10 ...
```

We then classify the variables according to their variable type, multilevel type, and whether they are essential in the main analysis, i.e., present in the model (see Table 1).

Table 1: Variables types in the impact data

Variable	Variable Type	Multilevel Type	Model type
id	ID	Sample unit	Auxiliary
name	ID	Cluster	Essential
type	categorical	Level-2	Essential
age	continuous	Level-1	Essential
motor_score	categorical	Level-1	Essential
pupil	categorical	Level-1	Essential
ct	categorical	Level-1	Essential
mort	binary	Level-1	Essential
hypox	binary	Level-1	Auxiliary
hypots	binary	Level-1	Auxiliary
tsha	binary	Level-1	Auxiliary
edh	binary	Level-1	Auxiliary

```
library("tidyverse")
dati <- dati|>
           mutate(across(-c("age","id","mort"), as.factor))
levels(dati$motor_score) <- c("1_2","3","4","5_6")
levels(dati$ct) <- c("I_II","III","IV_V")
```

With the **ggmice** function `plot_pattern()`, we can visualize the missing data pattern. The ten most frequent missingness patterns are shown with the following instruction:

```
library("ggmice")
plot_pattern(dati, rotate = TRUE, cluster= "name", npat = 10L)
```

By specifying the cluster parameter in the `plot_pattern` function, one can visualize not only the patterns but also which of them are more frequent across the clusters. For instance, the first, third and last patterns are present in all clusters, as indicated by the strong opacity, while the other patterns are only present in some clusters, as they are more transparent. Figure 2 shows that we need to impute the essential variables `ct` and `pupil`.

Particularly for clustered data, we need to examine whether the missingness is systematic or sporadic in the incomplete variables. To do this, we can calculate the percentage of missingness per group as follows:

```
library("pheatmap")
dati |>
  group_by(name)|>
  dplyr::summarize(across(everything(),
                          ~ (sum(is.na(.))*100/length(.))))|>
  tibble::column_to_rownames(var = "name")|>
  pheatmap::pheatmap(display_numbers = T,
                     cluster_rows=FALSE, cluster_cols=FALSE)
```

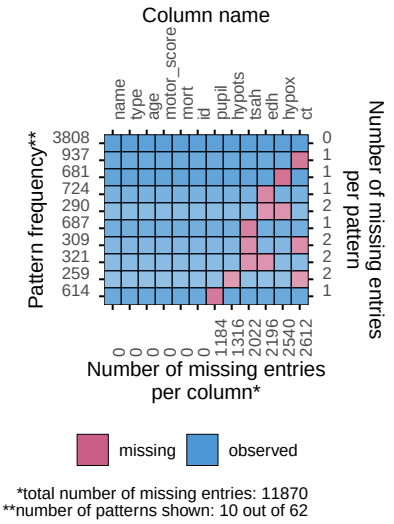


Figure 2: Missingness pattern in the impact data

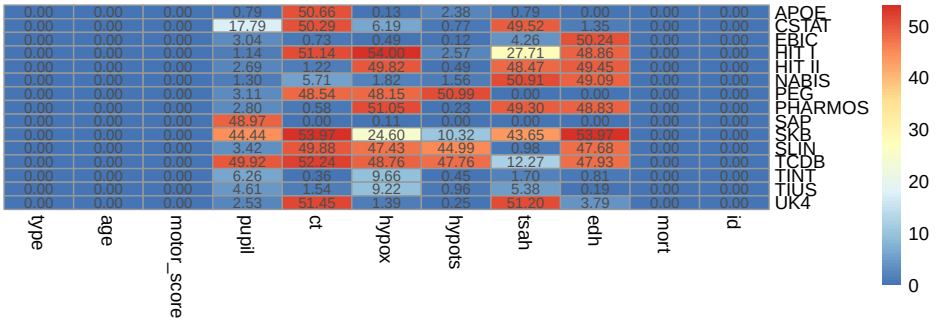


Figure 3: Missingness % per cluster in the impact dataset

From Figure 3, we observe that none of the variables is systematically missing but there are auxiliary variables that could serve to impute ct and pupil at the cluster level.

We can further evaluate which variables may be useful as auxiliaries by plotting the flux plot, which helps identify potential predictors of missing values among the auxiliary variables.

```
library(mice)
fluxplot(dati)
```

Figure 4 reveals that the majority of auxiliary variables have an outflux value ranging between 0.5 and 0.8, indicating that they are not powerful predictors (outflux > 0.9) but could still provide information for predicting other incomplete variables. Additionally, in this plot, we can observe that the 'hypox' and 'ct' variables have the highest influx, indicating that they are more dependent on the imputation model compared to other variables in the dataset.

We can also check the pairwise correlation plot as follows:

```
R> plot_corr(dati, rotate = TRUE)
```

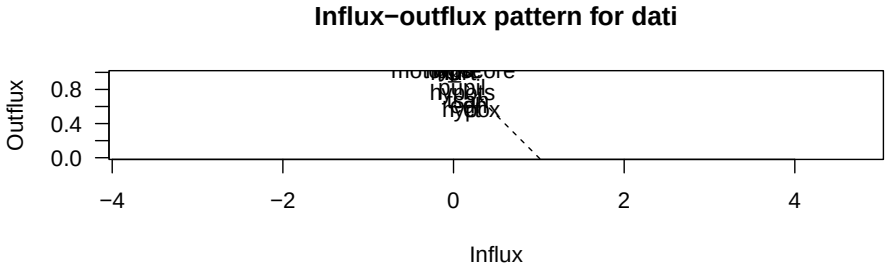


Figure 4: Influx-outflux pattern for the impact data

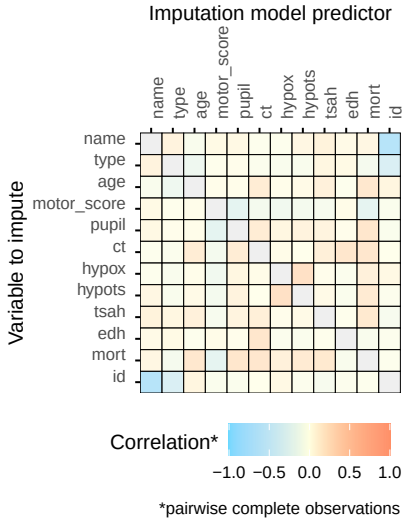


Figure 5: Pairwise correlation in the impact dataset

```
library(ggcorrplot)
model.matrix(~0+.,
             data=dati|>select(!name))|>
    cor(use="pairwise.complete.obs") |>
    ggcorrplot(show.diag=FALSE,
               type="lower",
               lab=TRUE,
               lab_size=2)
```

Figure 6 shows us that `hypox` and `hypot` would not be useful auxiliary variables for imputing `ct`. Depending on the minimum required correlation, `tsah` could be useful, while `edh` has the strongest correlation with `ct`(IV_V level) out of auxiliary variables in the data and should definitely be included in the imputation model. For the imputation of `pupil`, `hypots` and `hypox` could be used given the relative correlation with the pupil None level. We conclude that we will use all auxiliary variables in the imputation model.

Finally, we assess the heterogeneity of mortality across the studies using the ICC. We employ

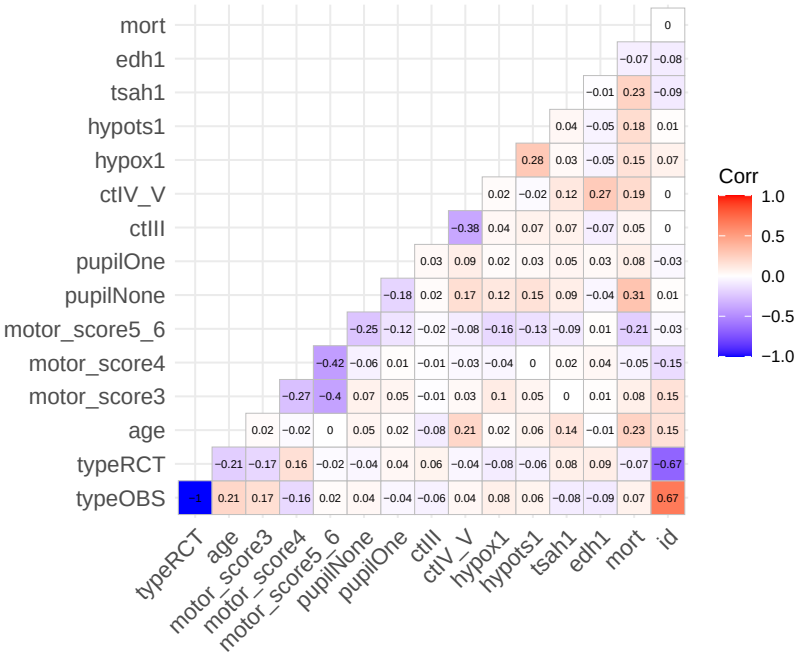


Figure 6: Pairwise correlation in the impact dataset

a random intercept model that incorporates the model predictors and level-2 variables (type of study) estimated on the complete observations.

```
mod_cc <- with(dati|> filter(complete.cases(dati)),
        lme4::glmer(mort~type+age+motor_score+ pupil+ct+(1|name),
        family = binomial))
performance::icc(mod_cc)

# Intraclass Correlation Coefficient

    Adjusted ICC: 0.023
  Unadjusted ICC: 0.016
```

The low ICC value (ICC=0.016) suggests that observations within clusters are not significantly more similar than observations across different studies. This indicates that a linear mixed model may not be necessary in the main model or for imputation, allowing us to proceed with a simpler strategy.

Imputation

Let's now proceed with the multiple imputation of this dataset. The first step is to set up the imputation method. We can use the `make.method` function, which suggests a plausible imputation method based on the type of variable. This function returns a vector specifying the imputation method for each variable in the dataset.

```
meth <- mice::make.method(dati)
meth
```

```
      name        type         age motor_score       pupil
        ""          ""          ""          ""   "polyreg"
        ct       hypox      hypots        tsah         edh
 "polyreg"    "logreg"    "logreg"    "logreg"    "logreg"
      mort          id
        ""          ""
```

By default, `mice` assigns the logreg (logistic regression) imputation method to binary variables, such as hypox and edh, and the polyreg (polytomous regression) imputation method to unordered factors with more than two levels, such as pupil and ct. It's worth noting that variables without any specified method, such as id and name, are complete. Here, we do not need to modify the default imputation methods, as we will follow the cluster indicator imputation strategy.

Next, we need to set the prediction matrix, we start by creating a predictor matrix using the `make.predictorMatrix` function. This results in a matrix where each row describes the prediction model of each incomplete variable on the dataset in terms of other variables arranged in the columns, where the cell value refers to the prediction code of the variable (See Box 4).

```
pred <- make.predictorMatrix(dati)
```

We can better visualize the prediction matrix and the imputation methods assigned to each incomplete variable by means of the `plot_pred` function from the **ggmice** package.

```
plot_pred(pred, method = meth, rotate = TRUE)
```

```
Ignoring unknown labels:
* colour : ""
```

The predictor matrix shows the variables that are included in the imputation model (prediction code 1) and those that are not included (prediction code 0). Additionally, the rows of the matrix only specify the imputation method for the incomplete variables in the dataset.

We still are required to modify the prediction matrix and remove some predictor variables. For instance, the "id" variable should not be used for imputation, as it has no predictive value for imputation, but we still we need to include the variable in the imputation dataset as afterwards it will be used for the identification of each observation.

Also, as differences between clusters will be explained solely by the cluster variable "name", the variations at level-2 are meaningless and should not be included in the imputation model. Therefore, we need to remove the level-2 variable "type", and the level-1 variable "id" as follows:

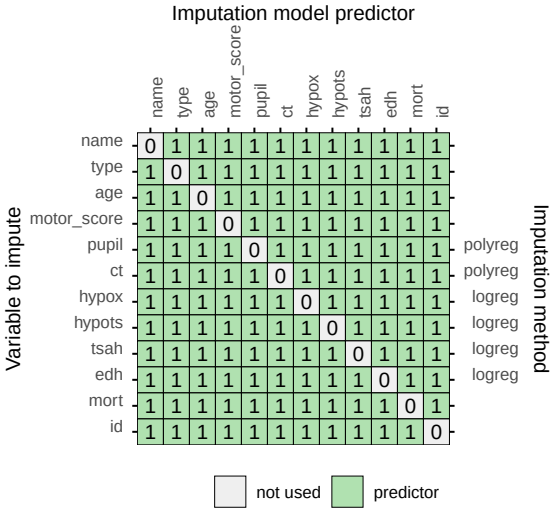


Figure 7: Prediction matrix in mice by ggmice

```
pred[ , c("id","type")] <- 0
```

At times, the predictor matrix can be constructed using the quickpred() function, which employs a variable selection procedure based on the correlation between the target missing variable and potential predictor variables in the dataset (Van Buuren 2018)(§ 9.1.6). However, it's crucial to verify the output of this function, to ensure that imputation models for the outcome variable do not lack essential variables or contain inconsistent multilevel terms, for instance, interaction terms without the inclusion of the main effects. To prevent essential variables from being excluded, include them with the "include" argument.

Then, we construct the 'visit' vector, to specify the order in which incomplete variables are imputed. Here, we ordered according to the missing percentage.

```
visit <- c("pupil","hypots","tsah","edh","hypox","ct")
```

Next, we proceed to impute the data by setting the `meth` and `pred` and `visit` parameters in the `mice` function. The output is a `mice` object that can be passed on to **mice** package functions and other package functions such as functions from the modeling packages **lme4** and **nlme**.

```
impcind <- mice(dati, pred = pred, meth = meth, visit = visit)
```

The cluster imputation strategy might suffice for the considered main model. The separate cluster imputation strategy might be applied also, but it might not be congenial with the main model as this imputation strategy assumes that β effects vary across clusters, which differs from the main model common effect.

If further complexities are taken into account in the study, such as allowing for random intercepts per study, that follow a Normal distribution, the main model would be represented instead as:

$$
\begin{aligned}
mort_{ij} &= Bernoulli(p_{ij}) \\
logit(p_{ij}) &= \alpha_j + \beta_1 age_{ij} + \beta_2 motor_score_{ij} + \beta_3 pupil_{ij} + \beta_4 ct_{ij} \\
\alpha_j &\sim N(\alpha, \tau^2) \qquad \text{(main model : 2)}
\end{aligned}
$$

where α is the overall intercept and τ^2 the between-study variance. In this scenario, the complexity of the imputation model should be increased by employing a 2l.model, thus facilitating a congenial model with the main model : 2 and avoiding bias in the variance components (Speidel *et al.* 2018). For brevity, the code for these three strategies is included in the Appendix, Section A2.

Post-imputation

After the imputation, we should check that the imputation model convereged, a necessary assumption for the validity of the imputation. For this, trace plots can be checked for each strategy. Here we show the plots for the cluster indicator strategy (Figure 8), which do not reveal any trend on the mean and standard deviation (sd) of the imputed variables. This is an indication of convergence, which allows us to continue.

```
plot_trace(impcind)
```

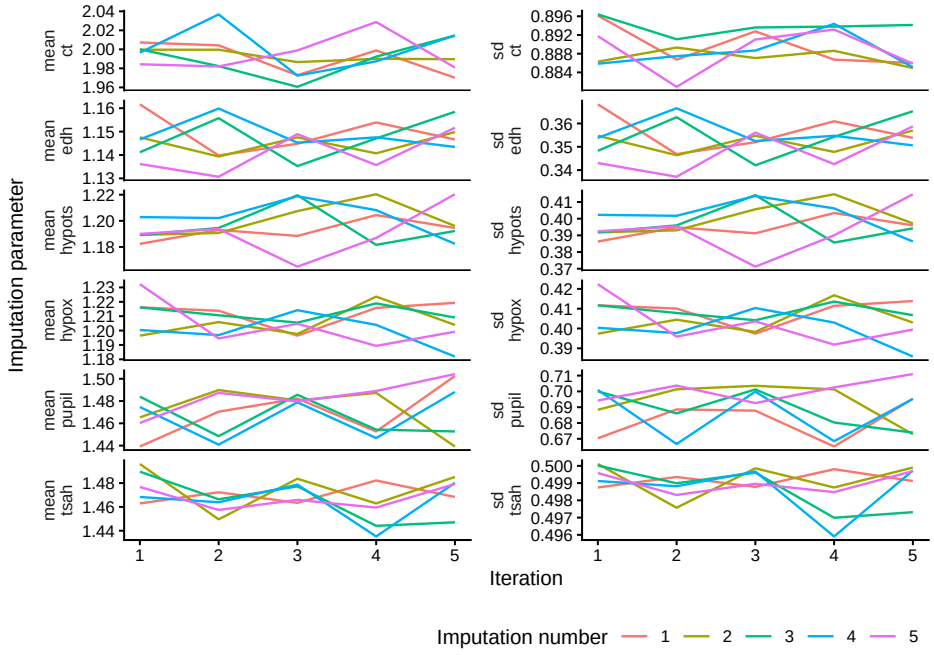
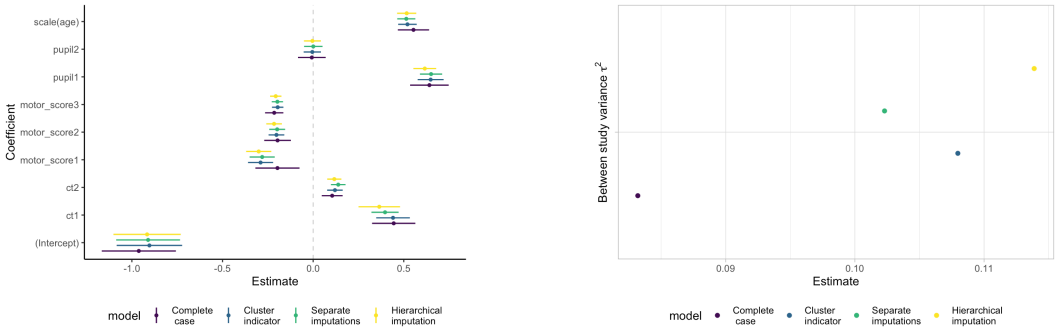


Figure 8: Trace plots for imputation with missing indicator strategy in the impact data

We estimate the random intercept model (main model : 2) using three imputation strategies along with a complete case analysis (CCA). Here, we display only the estimation results. In

the following section, we will detail the procedure of estimation and pooling of parameters from a multilevel model. The code to reproduce the figures is also included in the Appendix (Section A3).



(a) Estimated coefficients

(b) Between studies variance

Figure 9: Estimates of the main model in the impact dataset

From Figure 9a, we observe that the estimates of the coefficients (common effects) are quite similar across the three imputation strategies. On the other hand, the estimates from CCA are slightly different, particularly for the motor_score1 effect, which has large confidence intervals. Regarding the estimates of the between-study variance (Figure 9b, we did not find large differences across the different strategies' estimates. However, the estimate from the CCA is notably lower compared to the estimates from these strategies.

4.2. Illustration of one-stage imputation in the Popularity data

For the second study case, we focus on demonstrating the imputation of clustered data using 1-stage 2L methods, which are based on a (generalized) linear mixed model simultaneously estimated across clusters. We consider the simulated 'popmis' dataset, which includes pupils ($n = 2000$) clustered within schools ($N = 100$). The following variables are of primary interest:

- **pupil**, pupil identifier within the school.

- **school**, school identification number;
- **popular**, pupil popularity, self-rating between 0 and 10;
- **sex**, pupil sex with levels 0 = boy, 1 = girl;
- **texp**, teacher experience in years;
- **teachpop**, teacher popularity rate given by the pupils.

The objective of this study is to develop a prediction model for pupil popularity, that accounts for heterogeneity between schools. The main analysis of the dataset will employ a multilevel model, specifically a multivariable regression with random intercepts and slopes for the different schools, which estimates the popularity of the j-th student belonging to the i-th school based on the gender of the student, the teacher's experience, and the teacher's

popularity as follows:

$$popular_{ij} = (\beta_0 + b_{0i}) + \beta_1 sex_{ij} + \beta_2 texp_{ij} + (\beta_3 + b_{3i}) teachpop_{ij} + \beta_4 texp_{ij} * teachpop_{ij} + \epsilon_{ij}$$

where β_0,β_1,β_2,β_3 and β_4 are the fixed effects, b_{0i} is the random intercept effect and b_{3i} is the random slope effect of teacher's popularity distributed as $\begin{bmatrix} b_0 \\ b_3 \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_0^2 & \rho\tau_0\tau_3 \\ \rho\tau_0\tau_3 & \tau_3^2 \end{bmatrix}\right)$, τ_0^2 and τ_3^2 are respectively the intercept and slope variances and ρ the correlation of the random effects. The error term ϵ_{ij} is distributed as $\epsilon_{ij} \sim N(0, \sigma^2)$, with population variance σ^2.

This multilevel approach enables us to simultaneously consider individual-level factors (such as gender) and school-level factors (such as teacher experience), providing a comprehensive understanding of their influence and interactions on pupil popularity scores.

We load the data into the R environment and select the relevant variables which results in the following data structure.

```
data("popmis", package = "mice")
str(popmis)
```

```
'data.frame':   2000 obs. of  7 variables:
 $ pupil   : int  1 2 3 4 5 6 7 8 9 10 ...
 $ school  : int  1 1 1 1 1 1 1 1 1 1 ...
 $ popular : int  NA NA 7 NA NA 7 NA NA NA NA ...
 $ sex     : int  1 0 1 1 1 0 0 0 0 0 ...
 $ texp    : int  24 24 24 24 24 24 24 24 24 24 ...
 $ const   : int  1 1 1 1 1 1 1 1 1 1 ...
 $ teachpop: int  7 7 6 6 7 7 7 7 7 6 ...
```

We start by identifying the variables according to the variable type and multilevel type (Table 2).

Table 2: Types of variables in the popmis data

Variable	Variable Type	Multilevel Type
pupil	ID	Sample unit
school	ID	Cluster
popular	continuous	Level-1
sex	binary	Level-1
texp	continuous	Level-2
teachpop	continuous	Level-1

We then ensure that R recognizes which variables are factor and binary variables. Therefore, we include the sex variable as a factor variable.

```
popmis$sex <- as.factor(popmis$sex)
```

Then, to explore the missingness, we should look at the missing data pattern. With the **ggmice** function `plot_pattern`, we can visualize the missing data pattern, which shows that the missingness is univariate, as only the outcome variable is missing (Figure 10).

```
plot_pattern(popmis, rotate=T, cluster="school")
```

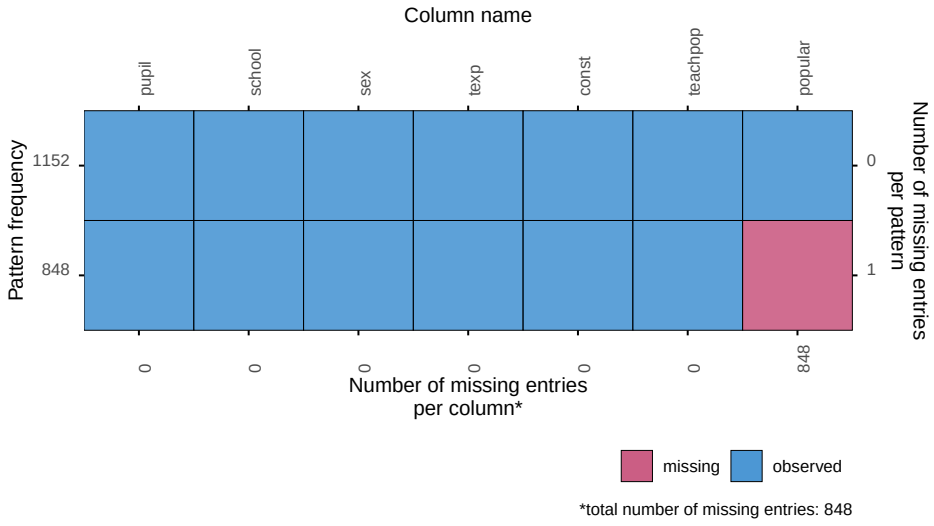


Figure 10: Missingness pattern in the popularity dataset

Given that the main analysis will use a multilevel model, and there are no additional predictor variables (plausible auxiliary variables) included in the dataset, in this case, the multiple imputation step will not be required, as the multilevel model is robust to missingness when only the outcome is incomplete (Molenberghs and Kenward 2007).

But for illustration purpose, we ampute some of the predictor variables (see the `ampute.popmis` function in the Appendix, Section A3.2}) and save the new incomplete data as the 'dat' object and proceed with the multiple imputation working flow.

```
popmis_amputed <- ampute.popmis(popmis)
```

```
plot_pattern(popmis_amputed, rotate = TRUE, cluster = "school")
```

The missing pattern in this dataset is depicted in Figure 11. Here we observe that the first two missingness patterns are present in all the clusters (i.e., schools), as indicated by the strong opacity, while the other patterns are only present in some schools, as they are more

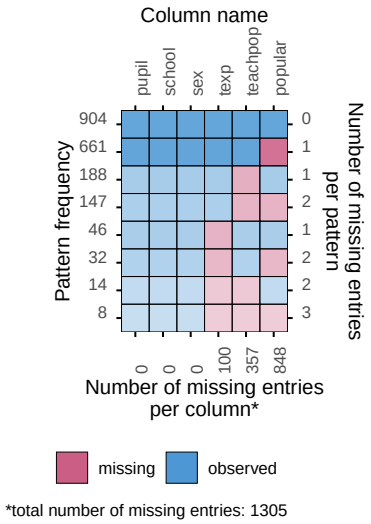


Figure 11: Missingness pattern in the amputed popularity dataset

transparent. Then, we examine whether the missingness of observations is systematic or sporadic across the included schools. To do this, we can check the percentage of missing values in the outcome variable across the clusters.

To develop an imputation model for the incomplete variable `popular`, we may check the association between the observed values of the target variable (in this case 'popular') and the values of other predictor variables and confirm if the multilevel structure considered in the main model is plausible (in this case a random slope on the `teachpop` variable). We can plot the association between popular and teacher popularity across the schools using observable data as follows:

```
ggplot(data = popmis_amputed,
   aes(x = teachpop, y = popular, col = school, group = school)) +
   geom_point(size=0.5, alpha=0.7, position="jitter") +
   theme_classic()+
   theme(legend.position="none")+
   scale_color_gradientn(colours = rainbow(100))+
   geom_smooth(method=lm, se=FALSE, linewidth=.5, alpha=.7)
```

Figure 13 shows that the effect of teacher popularity on the student popularity varies across the clusters, justifying the use of the random slope.

In addition, the estimated ICC of 0.25 in the complete dataset suggests that some variability in the data can be attributed to the cluster structure. Hence, a multilevel model is required for both the main analysis and the imputation model to ensure congeniality.

```
mod_cc <- nlme::lme(popular ~ sex + teachpop + texp + teachpop:texp,
            random =  ~ 1 + teachpop| school,
            data = popmis_amputed|>filter(complete.cases(popmis_amputed)),
            na.action = na.exclude)
```

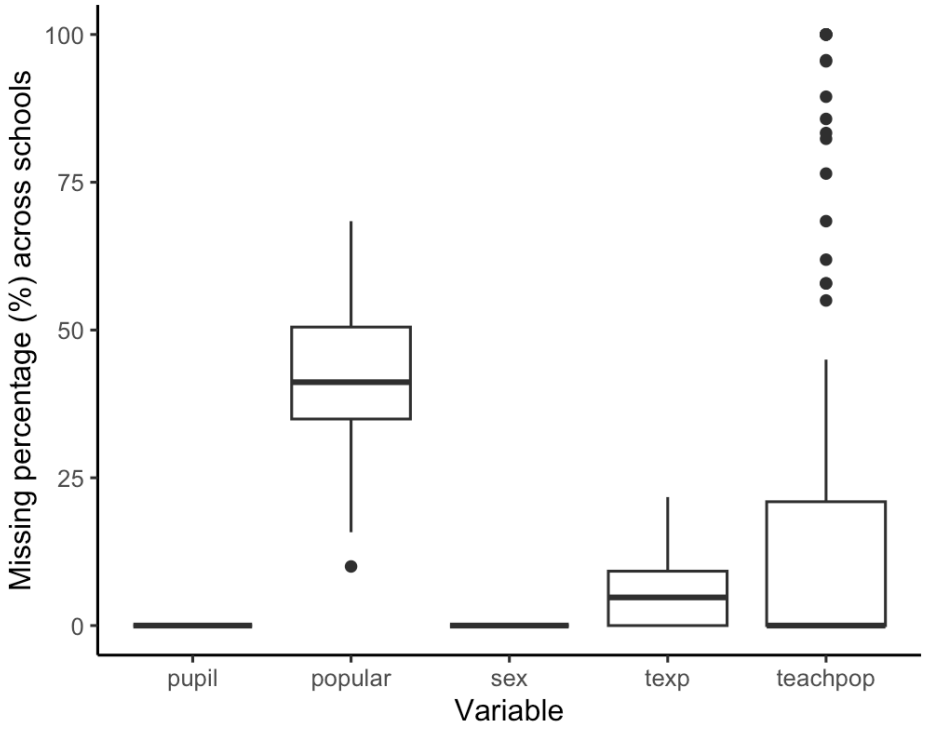


Figure 12: Missingness percentages per variable across the clusters

```
performance::icc(mod_cc)

# Intraclass Correlation Coefficient

    Adjusted ICC: 0.250
  Unadjusted ICC: 0.122
```

The association between incomplete variables and the predictors, for instance popularity and teacher experience, can be also visualized with **ggmice**. We can see the distribution of the teacher experience for cases where `popular` is observed and cases where `popular` is missing. It appears that students with a missing value for `popular` are in clusters with a slightly higher `texp` value (Figure 14).

```
ggmice(popmis_amputed, aes(texp)) +
   geom_density() +
   facet_wrap(~is.na(popular))
```

Finally, we need to prepare the dataset to be imputed, saved here as `dati`. As the main model includes a `texp*teachpop` interaction, we need to explicity add the `teachpop.texp` variable into the dataset without specifying any value.

```
popmis_amputed$teachpop.texp <- NA
```

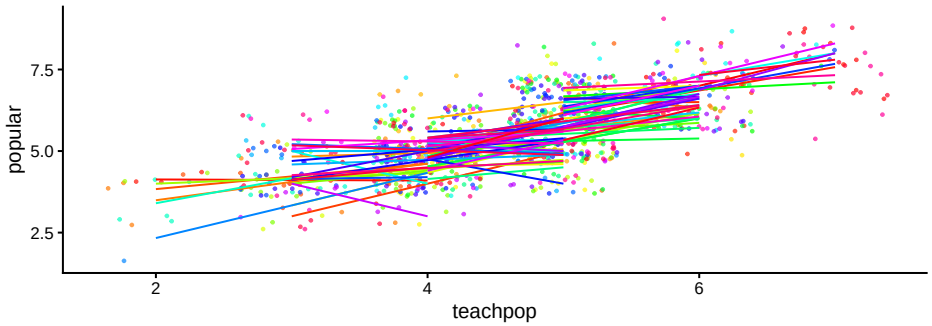


Figure 13: Association of popular and teacher popularity across clusters

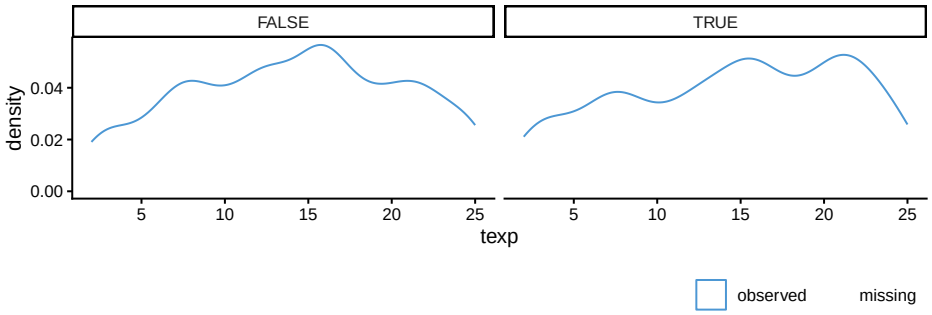


Figure 14: Density plot teacher experience according to the missingness of popular variable

Imputation

As before, we create the method vector with the `make.method` function.

```
meth <- mice::make.method(popmis_amputed)
meth
```

```
      pupil      school     popular         sex        texp
         ""          ""       "pmm"          ""       "pmm"
teachpop.texp
    "logreg"
```

This function suggests imputing all the incomplete variables using the predictive mean matching method (pmm), except for the interaction term 'teachpop.texp'. However, this function does not support 2l.methods yet. Therefore, manual modification of the methods is necessary. For the level-1 variables (popular, teachpop), we can use the '2l.pmm' method from the `miceadds` package. For the level-2 variable (texp), the '2lonly.mean' method can be applied due to sporadic missingness within clusters. Finally, for the 'teachpop.texp' variable, specifying the interaction can be achieved using the I notation as follows:

```
library("miceadds")
meth[c("popular","teachpop")] <- "2l.pmm"
```

```
meth["texp"] <- "2lonly.mean"
meth[c("teachpop.texp")] <- "I(teachpop*texp)"
```

Next, we need to set the prediction matrix. Here we follow the recipes for multilevel datasets provided by (Van Buuren 2018) ($§§7.10) as the main model involves interaction at different levels. Therefore, we start by creating a predictor matrix using the make.predictor function. Initially, we will set a blank matrix where no predictor is assigned in any of the prediction models (denoted as prediction code = 0, see Box 4). Then we proceed to specify the cluster variable (prediction code = -2) across all the imputation models.

```
pred <- make.predictorMatrix(popmis_amputed)
pred[,] <- 0
pred[,"school"] <- -2
```

We define the imputation model of the outcome using the main model. Specifically, we include the 'texp' level-2 variable as a fixed effect (prediction code = 1). The 'sex' variable is also included as a fixed effect in the main model. As suggested by recipes ((Van Buuren 2018)), include the group mean of the variable (prediction code = 3) to avoid convergence problems. Additionally, we include the 'teachpop' variable, which requires a random slope in the main model, along with its group mean (prediction code = 4). Finally, we add the cross-interaction term as a fixed effect (prediction code = 1).

```
pred["popular",c("sex","texp","teachpop","teachpop.texp")] <- c(3,1,4,1)
```

The imputation model of the interaction terms is specified with the variables used for its deterministic imputation:

```
pred["teachpop.texp",c("texp","teachpop")] <- c(1,1)
```

Additionally, we define the imputation model for other incomplete variables by incorporating the group mean and avoiding circular dependencies. For example, we refrain from including predictor variables in the imputation model that were deterministically imputed with the incomplete variable under consideration. In essence, we avoid using interaction terms such as `teachpop.texp` in the imputation model for `teachpop`, as they are defined by the same variable.

```
pred["teachpop", c("popular","sex","texp")] <- c(4,3,1)
```

Following the recipes, we should also include the interaction between the group mean of teachpop and texp. However, when we tried this, convergence issues became evident, as indicated by the trace plot (not shown). Consequently, we decided to exclude this interaction term from the imputation model.

We can visualize the adjustments in the imputation methods and predictors by means of the `plot_pred` **ggmice** function (Figure 15).

```
plot_pred(pred, method = meth, rotate = TRUE) + guides(fill="none")
```

```
Ignoring unknown labels:
* colour : ""
```

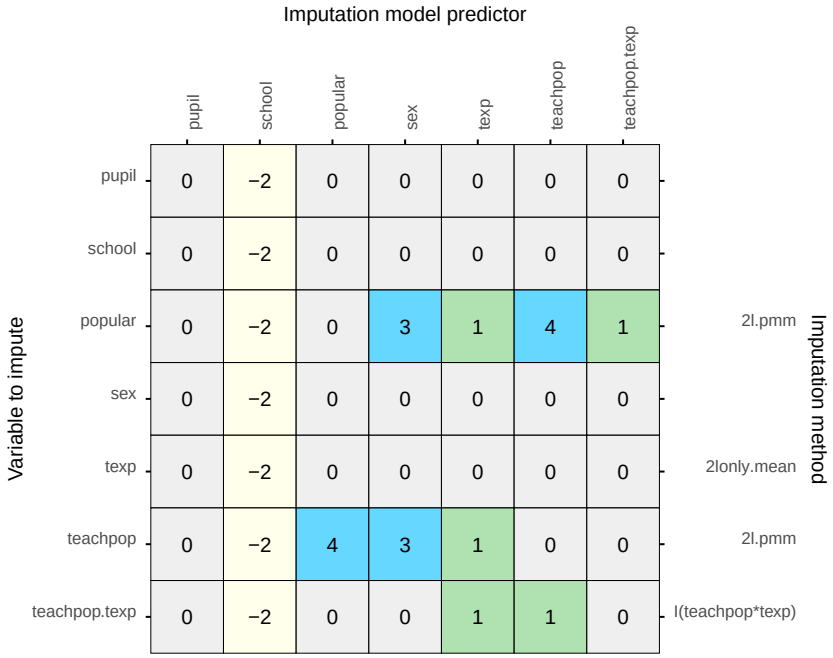


Figure 15: Imputation predictor matrix for the popularity dataset

Finally, we proceed to run the imputation in `mice`. For the imputation order we ensure that the interaction for `texp` and `teachpop` is imputed immediately after its individual components, to ensure consistency when imputing the remaining variables. Further, we order by the percentage of missingness.

```
imp_popmis <- mice(data = popmis_amputed,
             method = meth,
             predictorMatrix = pred,
             visit = c("texp","teachpop","teachpop.texp","popular"),
             printFlag = FALSE)
```

Now we need to check the convergence of the imputation procedure. This can be done, for instance, by examining the trace plots. The trace plot (Figure 16) does not reveal any convergence problems, as no discernible patterns are observed in the mean and standard deviation of the popularity variable.

```
plot_trace(imp_popmis)
```

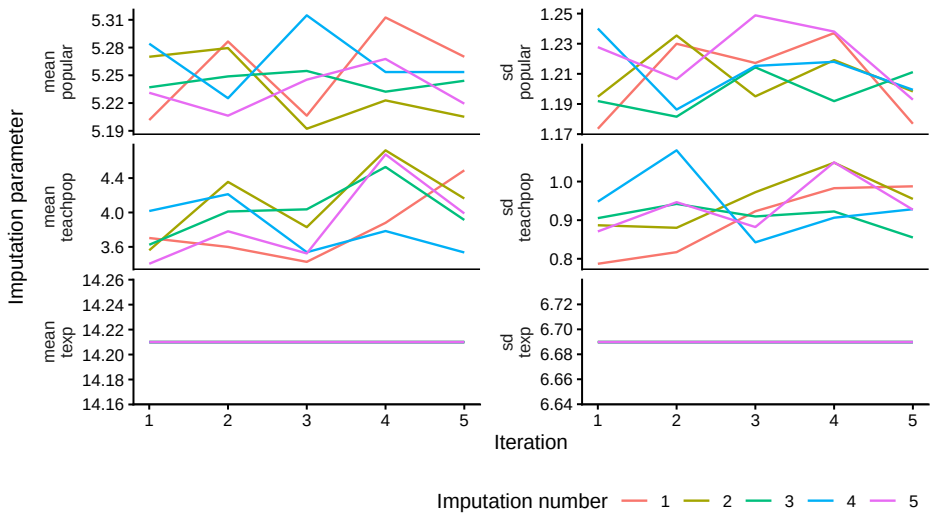


Figure 16: Trace plot of imputed popularity dataset

```
Trace plot could not be produced for variable(s):
  pupil, school, sex, teachpop.texp
i No convergence diagnostics found.
```

We can also evaluate the distribution of the imputed values and compare it to the distribution of the observed values as follows, where we observe that imputed values are in line with the observed ones (Figure 17):

```
ggmice(imp_popmis,
   aes(popular, group = .imp)) +
   geom_density()
```

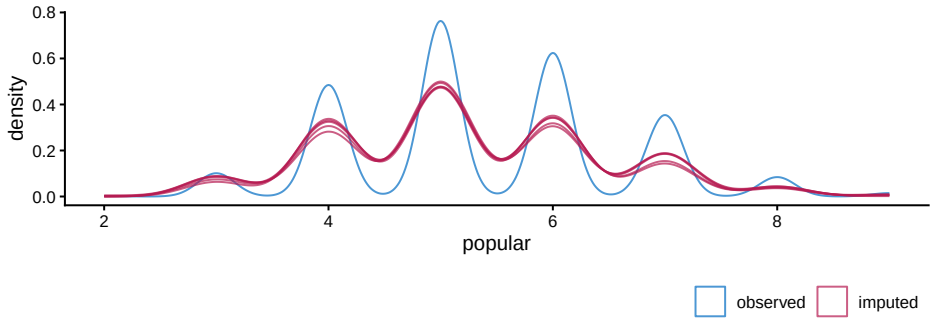


Figure 17: Density plot of imputed popularity values

Finally, we can estimate the main analysis model on the imputed datasets, by simply passing the `mice` output object (imp_popmis) to the **lme** function from the **nlme** package using the `with` function.

```
fit_popmis <- with(imp_popmis,
   nlme::lme(popular ~ sex + teachpop + texp+ teachpop:texp,
   random =  ~ 1 + teachpop| school))
```

We can pool the results using the `pool` function from the **broom.mixed** package. However, this option only allows pooling of coefficients (parameters of fixed effects). Alternatively, we can utilize the **testEstimates** function from the **mitml** package, which enables pooling not only from coefficients, but also from variances of the random effects, as follows:

```
library("mitml")
est_popmis <- testEstimates(as.mitml.result(fit_popmis),
                            extra.pars = TRUE)
```

The fixed coefficients can be extracted directly with the `estimates` parameter:

```
library("knitr")
est_popmis$estimates |>
   round(3) |>
   kable()
```

	Estimate	Std.Error	t.value	df	P(>|t|)	RIV	FMI
(Intercept)	2.926	0.312	9.377	197.350	0.000	0.166	0.151
sex1	0.771	0.054	14.283	8.995	0.000	2.002	0.722
teachpop	0.238	0.068	3.503	168.435	0.001	0.182	0.164
texp	0.032	0.024	1.370	53.746	0.176	0.375	0.298
teachpop:texp	0.007	0.004	1.636	591.775	0.102	0.090	0.085

and the variances of the model with the `extra.pars` parameter. This provides the estimates for the variance of the random intercept, variance of the random slope of the `teachpop` variable, the covariance between the intercept and random slope, and the ICC, respectively, as follows:

```
est_popmis$extra.pars |>
     round(3) |>
     kable()
```

	Estimate
Intercept~~Intercept|school	0.476
teachpop~~teachpop|school	0.016
Intercept~~teachpop|school	-0.064
Residual~~Residual	0.446
ICC|school	0.490

Estimate

4.3. Illustration of 2-stage imputation in the Obesity data

In the third case study, we demonstrate a multilevel imputation using a random intercept and random slope model by using the 2-stage 2L imputation methods, where separate imputation models are fitted across the clusters and then combined using meta-analysis techiques.

We employ the obesity dataset included in the **micemd** package, a simulated dataset that emulates an electronic survey in which individuals are asked to provide information about their weight and consumption habits in different countries. We use the following variables from the dataset:

- **Cluster**: region of the patients' healthcare provider,
- **Gender**: subjects' Gender (0=male, 1=female),
- **Age**: subjects' age,
- **Height**: subjects' height in metres,
- **Weight**: subjects' weight in kilograms,
- **BMI**: subjects' body mass index,
- **FamOb**: family obesity history (yes or no),
- **Time**: response time in minutes (exclusion-restriction variable).

In this dataset, `Age` and `FamOb` are MAR, while the `Weight` variable is affected by selection bias, which we assume to be caused by an indirect MNAR mechanism. This MNAR mechanism typically arises when an unobserved or omitted variable influences both the value of the incomplete variable (in this case, `Weight`) and its likelihood of being missing.

In the main analysis model, BMI serves as the dependent variable, with `Age`, `Gender`, and `FamOb` as predictors. Because of the clustered nature of the data, we include random intercepts, as well as random slopes for the `Age` variable. Therefore, the model for the i-th region and the j-th subject is represented as:

$$BMI_{ij} = (\beta_o + b_{oj}) + (\beta_1 + b_{1j}) * Age_{ij} + \beta_2 * FamOb_{ij} + \beta_3 Gender_{ij} + \epsilon_{ij} \qquad (1)$$

where β_0, β_1, β_2, and β_3 are the fixed effects, b_{0i} and b_{1i} are the random intercept and age slope effects distributed as $\begin{bmatrix} b_0 \\ b_1 \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_0^2 & \rho\tau_0\tau_1 \\ \rho\tau_0\tau_1 & \tau_1^2 \end{bmatrix}\right)$, where we denote τ_0^2 and τ_1^2 for the intercept and slope variances and ρ the correlation of the random effects. The error term ϵ_{ij} is normally distributed as $\epsilon_{ij} \sim N(0, \sigma^2)$, with variance σ^2.

Pre-imputation

We start by loading the data and check the structure.

```
data(Obesity, package = "micemd")
Obesity <- as.data.frame(Obesity)
str(Obesity)
```

```
'data.frame':   2111 obs. of  8 variables:
 $ Cluster: int  1 1 1 1 1 1 1 1 1 1 ...
 $ Time   : num  5.54 3.09 5.74 1.99 3.74 ...
 $ Gender : Factor w/ 2 levels "Female","Male": 1 1 2 1 1 1 1 2 1 2 ...
 $ Height : num  1.69 1.77 1.6 1.68 1.75 ...
 $ FamOb  : Factor w/ 2 levels "no","yes": 2 2 2 2 2 1 1 2 2 2 ...
 $ Age    : num  20 29 30 24 26 18 27 20 29 19 ...
 $ Weight : num  109.9 100.1 95.7 94.5 101.2 ...
 $ BMI    : num  38.4 32 37.3 33.5 33 ...
 - attr(*, ".internal.selfref")=<pointer: (nil)>
```

We then classify each variable according to variable type, multilevel type and if they are essential in the model (Table 5). Weight and Height are considered essential even when they are not formally included in the model as BMI is deterministically obtained from them.

Table 5: Variable types of the Obesity data

Variable	Variable Type	Multilevel Type	Model type
*	Subject ID	Sample unit	Auxiliary
Cluster	ID	Cluster	Essential
Gender	binary	Level-1	Essential
FamOb	continuous	Level-1	Essential
Age	continuous	Level-1	Essential
Weight	continuous	Level-1	Essential
Height	continuous	Level-1	Essential
Time	continuous	Level-1	Auxiliary

*Not included in the data, but included in this table for clarity.

Next, we examine the observable data and evaluate the ICC on the complete observations.

```
mod_cc <- with(Obesity|>filter(complete.cases(Obesity)),
               nlme::lme(BMI ~ Age + FamOb + Gender,
                         random =  ~ 1 + Age | Cluster, keep.data = TRUE))
performance::icc(mod_cc)
```

```
# Intraclass Correlation Coefficient

    Adjusted ICC: 0.383
  Unadjusted ICC: 0.374
```

The estimated ICC equals 0.383, which suggests that we require of a imputation method that accounts for clustering.

By plotting `BMI` against `Age` (see Figure 18). From the plot, it is evident that incorporating a random slope for `Age` might be beneficial for the main model and the imputation model.

```
ggplot(data = Obesity,
          aes(x = Age, y = BMI, col = Cluster, group = Cluster)) +
          geom_point(size=0.5, alpha=0.7, position="jitter") +
          theme_classic()+
          scale_color_gradientn(colours = rainbow(5))+
          geom_smooth(method=lm, se=FALSE, size=.5, alpha=.7)+
          facet_grid(~Cluster )+
          theme(legend.position="none")
```

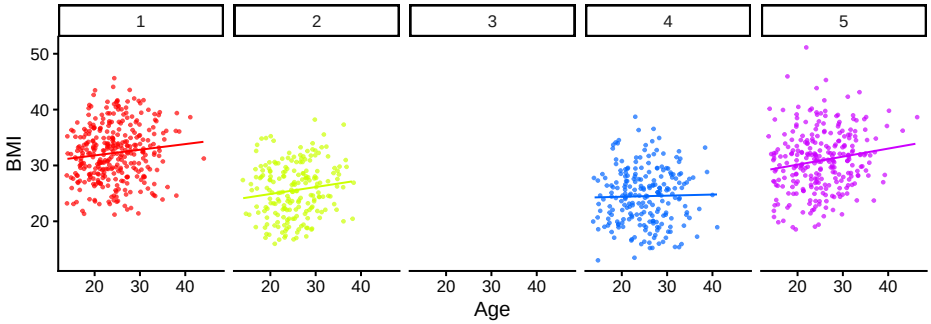


Figure 18: BMI vs Age in the observed data of the Obesity dataset

Now, let's examine the missingness patterns and the missingness percentage in the data by cluster:

```
plot_pattern(Obesity, rotate = T, cluster = "Cluster")
```

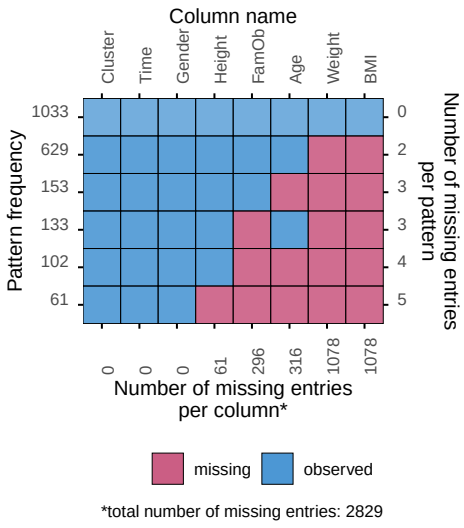


Figure 19: Missingness exploration in the obesity dataset: missingness pattern

```
Obesity|>
    group_by(Cluster)|>
    dplyr::summarize(across(everything(),~ (sum(is.na(.))*100/
    length(.))))|>
    tibble::column_to_rownames(var = "Cluster")|>
```

```
pheatmap::pheatmap(display_numbers = T,
cluster_rows=FALSE, cluster_cols=FALSE)
```

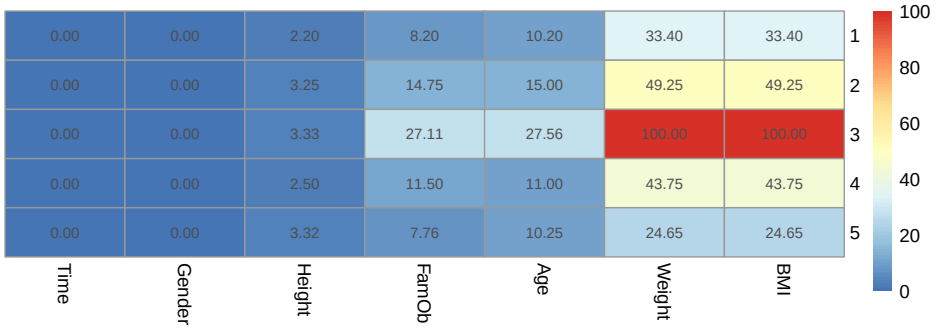


Figure 20: Missingness exploration in the obesity dataset: missingness percentage

We observe that the missingness pattern is non-monotonic and quite similar across the clusters. However, we notice that the `Weight` and `BMI` variables are systematically missing in cluster 3 (Figures 19 and 20).

Imputation

In this example, we opt to use the 2L imputation methods from the **micemd** package, which facilitate the imputation of incomplete variables under sporadic and systematic missingness. Within this package, various methods are available, including the 2l.stage methods (Audigier *et al.* 2018), where model estimation is conducted in two stages.

This package provides the `find.defaultMethod` function, which recommends a suitable imputation method for MAR variables based on variable type, cluster size, and number of clusters. Here, we apply this function to provide an imputation method for all incomplete variables, including the cluster variable position in the dataset (`ind.clust = 1`) and using the function's default parameters.

```
library("micemd")
meth_mar <- find.defaultMethod(Obesity,
                               ind.clust = 1, I.small = 7,
                               ni.small = 100, prop.small = 0.4)
```

It suggests using '2l.2stage.bin' for the FamOb variable and '2l.2stage.norm' for the `Age` variable. However, upon inspecting the density plot for `Age`, we found the variable to be highly skewed, prompting us to consider modifying its method to '2l.2stage.pmm'. For the BMI variable, we employ deterministic imputation.

```
meth_mar["Age"] <- "2l.2stage.pmm"
meth_mar["Weight"] <- "2l.2stage.pmm"
meth_mar["BMI"] <- "~I(Weight/(Height)^2)"
```

For these imputation methods, it is necessary to specify a prediction matrix, with the cluster variable labelled as -2. However, here the recipe that we used before is not longer necessary as these methods assume that all the imputation models contains also a random slope effect, so they use imputation Code 2 instead. We need to suppress the variable `Time` as this variable is not specified in the main model. We also modify the relationship between `BMI`, `Weight` and `Height` in the prediction matrix to avoid circularity issues.

```
pred_mar <- make.predictorMatrix(Obesity)
pred_mar[, "Time"] <- 0
pred_mar[, "Cluster"] <- -2
pred_mar[pred_mar == 1] <- 2
pred_mar[c("Height", "Weight"), "BMI"] <- 0
```

Then we proceed to run the imputation model, using as visit order the variables order according to their missing percentage.

```
imp_mar <- mice(data = Obesity,
                method = meth_mar,
                predictorMatrix = pred_mar,
                visit = c("Height","FamOb","Age","Weight","BMI"),
                printFlag = FALSE)
```

We consider the possibility that patients may not have been selected randomly, which would then have led to a distribution for weight that does not reflect the weight in the population. It's likely that an omitted variable, like self-esteem, could influence this selection. For example, individuals with lower self-esteem might have higher weight values, which would affect their willingness to provide honest information as they feel embarrassed.

To address this situation where data are missing not at random (MNAR), one approach is to apply the Heckman selection model (Muñoz *et al.* 2023a). This method has recently been extended to allow for variations in intercepts and exposure effects (random intercept and slope) (Galimard, Chevret, Curis, and Resche-Rigon 2018; Muñoz, Hufstedler, Gustafson, Bärnighausen, De Jong, and Debray 2023b). This method involves specifying two equations: one for the outcome, describing the incomplete variable in terms of partially observed predictors (in this case, all variables from the main model), and the other for the selection model, explaining the probability of being observed based (denoted here as R) on certain variables.

With this method, the `Weight` variable should be specified as `2l.2stage.heckman`.

```
meth_mnar <- meth_mar
meth_mnar["Weight"] <- "2l.2stage.heckman"
```

Additionally, the prediction matrix needs modification, as this model includes one submodel for the individuals' measured outcome, and one for the selection of individuals.

For the outcome equation we consider the same imputation model that we used for the MAR case (main model).

$$Weight_{ij} = \beta_o^O + \beta_1^O Age_{ij} + \beta_2^O FamOb_{ij} + \beta_3^O Gender_{ij} + \epsilon_{ij}^O$$

Regarding the selection equation, we include the same predictors as those in the main model, as well as a time variable. Here the time variable serves as a restriction exclusion variable specifically explaining the probability of being observed but not affecting the incomplete value (Weight). In this context, we assume that the time a user spends completing the survey serves as a proxy for the barriers they may encounter in survey completion, such as familiarity with the survey content or internet speed. These factors may lead the user to skip specific questions or even the entire survey. Also, we assume the time does not have any influence on the subject's weight.

$$R_{ij} = \beta_o^S + \beta_1^S Age_{ij} + \beta_2^S FamOb_{ij} + \beta_3^S Gender_{ij} + \beta_4^S Time_{ij} + \epsilon_{ij}^S$$

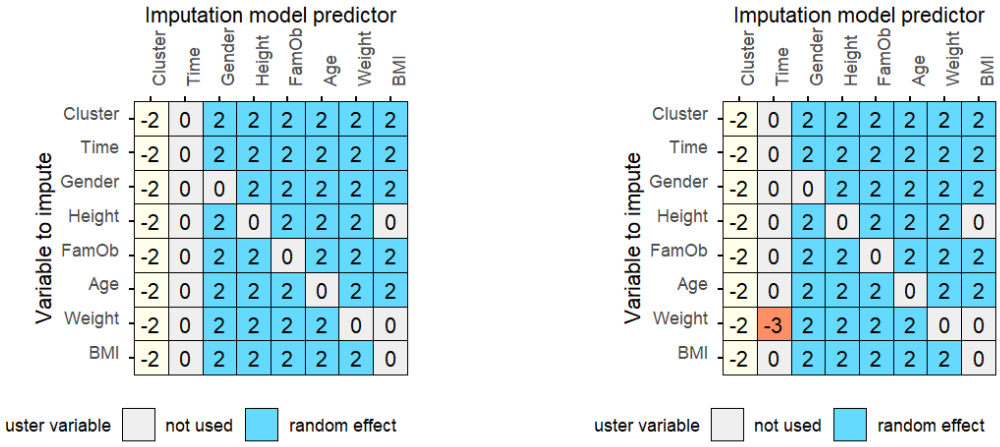
These two equations are jointly estimated under the assumption that the error terms are interconnected with a bivariate normal distribution. For a more comprehensive understanding of the model and the exclusion restriction, see (Muñoz *et al.* 2023b).

To use information from both equations, we must adjust the prediction matrix. The cluster variable remains specified as before (-2). In this imputation method, all the variables present in both the selection and outcome equations are included with a random effect.

However, it is essential to distinguish which of these variables appear in each equation. In this framework, when a variable is shared between both equations, it is denoted as (2). Predictors exclusive to the outcome equation are indicated as (-4), while those exclusive to the selection equation are labelled as (-3). Consequently, the only alteration needed in the predictor matrix pertains to the variable `Time`. Predictor matrices of both scenarios (MAR and MNAR) are displayed in Figure 21:

```
pred_mnar <- pred_mar
pred_mnar["Weight","Time"] <- -3

plot_pred(pred_mar, rotate = TRUE)
plot_pred(pred_mnar, rotate = TRUE)
```



(a) MAR (b) MNAR

Figure 21: Predictor matrices for the MAR and MNAR imputation models in the Obesity dataset

As in the MAR assumption, we use the 'pmm' option but this time for the Heckman imputation. This approach ensures that the imputed values remain within the range of observable values. We then run the MNAR imputation model by setting the `pmm` parameter to true.

```
# copulaIPD <- function (data, sel, out, family, send) {
#   fit_ind <- 0
#   fit <- try(GJRM::gjrm(formula = list(sel, out),
#                         data = data,
#                         margins = c("probit", ifelse(family == "binomial", "probit", "N")),
#                         model = "BSS",
#                         parscale = TRUE),
#              silent = TRUE)
#   if (!any(inherits(fit, "try-error"))) {
#     ev <- eigen(fit$fit$hessian, symmetric = TRUE, only.values = TRUE)$values
#     convh <- min(ev) > 0
#     convg <- max(abs(fit$fit$gradient)) < 10
#     CIcon <- summary(fit)$CItheta
#     MNAR_ind <- !(CIcon[[1]] > -0.001 & CIcon[[2]] < 0.001)
#     if (MNAR_ind) {
#       fit_ind <- 2
#       if (convh & convg) {
#         fit_ind <- 1
#       }
#     }
#   }
#   if (fit_ind == 2) {
#     fit <- NULL
#     gam1 <- try(mgcv::gam(formula = sel, data = data, family = "binomial",
```

```
#                           method = "REML"))
#     gam2 <- try(mgcv::gam(formula = out, data = data, family = family,
#                           method = "REML"))
#     fit_ind <- 0
#     if (!any(inherits(gam1, "try-error")) & !any(inherits(gam2,
#                                                           "try-error"))) {
#       coefficients <- c(gam1$coefficients, gam2$coefficients)
#       vp <- c(diag(gam1$Vp), diag(gam2$Vp))
#       if (all(c(!is.na(coefficients), vp != 0))) {
#         s <- ifelse(family != "binomial", 1, 0)
#         ncol1 <- ncol(gam1$Vp)
#         ncol2 <- ncol(gam2$Vp)
#         Vb <- matrix(0, ncol = ncol1 + ncol2 + 1 + s,
#                      nrow = ncol1 + ncol2 + 1 + s)
#         Vb[1:ncol1, 1:ncol1] <- gam1$Vp
#         Vb[(ncol1 + 1):(ncol1 + ncol2), (ncol1 + 1):(ncol1 +
#                                                      ncol2)] <- gam2$Vp
#         Vb[nrow(Vb), nrow(Vb)] <- 1/nrow(data)
#         if (family != "binomial") {
#           coefficients <- c(coefficients, sigma.star = log(sqrt(gam2$scale)))
#           Vb[(nrow(Vb) - 1), (nrow(Vb) - 1)] <- gam2$V.sp
#         }
#         fit$coefficients <- c(coefficients, theta.star = 0)
#         fit$Vb <- Vb
#         fit_ind <- 2
#       }
#     }
#   }
#   if (fit_ind != 0) {
#     names <- c(paste0(names(fit$coefficients)[1:send], "_s"),
#                names(fit$coefficients[(send + 1):length(names(fit$coefficients))]))
#     names(fit$coefficients) <- names
#     colnames(fit$Vb) <- names
#     rownames(fit$Vb) <- names
#   }
#   else {
#     fit <- NA
#   }
#   return(list(fit, fit_ind))
# }
#
# mice.impute.2l.2stage.heckman.valentijn <-
# function (y, ry, x, wy = NULL, type, pmm = FALSE, ypmm = NULL,
#     meta_method = "reml", pred_std = FALSE, ...) {
#     if (pred_std == TRUE) {
#         x <- apply(x, MARGIN = 2, FUN = function(X) (if (length(unique(X)) <=
#             2 & sum(unique(X)) == 1) {
#             X
#         }
#         else {
#             scale(X)
#         }))
#     }
#     colnames(x) <- paste0("x_", 1:length(colnames(x)))
#     bos_name <- colnames(x)[type == 2]
#     sel_name <- colnames(x)[type == -3]
#     out_name <- colnames(x)[type == -4]
```

```
#     if ((inherits(y, "factor")) & nlevels(y) == 2) {
#         family <- "binomial"
#     }
#     else {
#         family <- "gaussian"
#     }
#     if (length(colnames(x)[type == -2]) == 0) {
#         message("No group variable has been provided, the Heckman imputation model will be applied globally
#         Grp_est <- 0
#     }
#     else {
#         group_name <- colnames(x)[type == -2]
#         names.clust <- as.character(unique(x[, group_name]))
#         Grp_est <- 1
#     }
#     order <- c(sel_name, bos_name, out_name)
#     send <- length(sel_name) + length(bos_name) + 1
#     oend <- send + length(bos_name) + length(out_name) + 1
#     out <- as.formula(paste0("y", "~", paste(c(bos_name, out_name), collapse = "+")))
#     sel <- as.formula(paste0("ry", "~", paste(c(sel_name, bos_name), collapse = "+")))
#     data <- data.frame(ry, y, x[, order])
#     X <- data.frame(cbind(Int = rep(1, nrow(x)), x[, order]))
#     Syst_nest <- Heck_est <- 1
#     if (Grp_est == 1) {
#         df_list <- split(data, x[, group_name])
#         res_total <- suppressWarnings(lapply(df_list, copulaIPD,
#             sel = sel, out = out, send = send, family = family))
#         studytype <- sapply(res_total, function(x) x[[2]])
#         fit_list <- lapply(res_total, function(x) x[[1]])
#         coef_list <- lapply(fit_list[studytype != 0], `[[`, c("coefficients"))
#         Vb_list <- lapply(fit_list[studytype != 0], `[[`, c("Vb"))
#
#         coef_mat_s <- do.call(rbind, coef_list)
#         varnam <- colnames(coef_mat_s)
#         selnam <- varnam[grepl("*_s", varnam)]
#         outnam <- varnam[!varnam %in% c(selnam, "sigma.star",
#             "theta.star")]
#         if (length(studytype[studytype != 0]) < 2) {
#             Grp_est <- 0
#         }
#         else {
#             Heck_mod <- micemd:::get_marginal(coef_mat_s, Vb_list, selnam, outnam, meta_method)
#             Heck_est <- Heck_mod$Mvma_est
#         }
#         Sys.nest <- as.numeric(length(studytype[studytype ==
#             0]) == 0)
#     }
#     if (Grp_est == 0 | Heck_est == 0) {
#         message("The Heckman model cannot be estimated marginally, so systematically missing groups will be
#         Heck_mod <- copulaIPD(data = data, sel = sel, out = out,
#             family = family, send = send)
#         if (Heck_mod[[2]] == 0 & (Grp_est == 0 | Syst_nest ==
#             0)) {
#             stop("There is insufficient information to impute the Heckman model at the marginal or study le
#         }
#     }
#     if (Grp_est == 1) {
```

```
#         for (i in names.clust) {
#             if (studytype[[i]] == 0) {
#                 star <- micemd:::star_systematic(Heck_mod, send, oend,
#                   family)
#             }
#             else {
#                 star <- micemd:::star_sporadic(Heck_mod, coef_list_i = coef_list[[i]],
#                   Vb_list_i = Vb_list[[i]], selnam, outnam, family)
#             }
#             Xm <- X[!ry & x[, group_name] == as.numeric(i), ]
#             if (nrow(Xm) != 0) {
#                 y.star <- micemd:::gen_y_star(Xm = Xm, sel_name = sel_name,
#                   bos_name = bos_name, out_name = out_name, beta_s_star = star$beta_s_star,
#                   beta_o_star = star$beta_o_star, sigma_star = star$sigma_star,
#                   rho_star = star$rho_star, pmm = pmm, ypmm = ypmm,
#                   y = y, ry = ry)
#                 y[!ry & x[, group_name] == as.numeric(i)] <- y.star
#             }
#         }
#     }
#     else {
#         star <- micemd:::star_systematic(Heck_mod, send, oend, family)
#         Xm <- X[!ry, ]
#         if (nrow(Xm) != 0) {
#             y.star <- micemd:::gen_y_star(Xm = Xm, sel_name = sel_name,
#                 bos_name = bos_name, out_name = out_name, beta_s_star = star$beta_s_star,
#                 beta_o_star = star$beta_o_star, sigma_star = star$sigma_star,
#                 rho_star = star$rho_star, pmm = pmm, ypmm = ypmm,
#                 y = y, ry = ry)
#             y[!ry] <- y.star
#         }
#     }
#     return(y[!ry])
# }


# meth_mnar <- meth_mar
# meth_mnar["Weight"] <- "2l.2stage.heckman.valentijn"
#
#
# imp_mnar <- mice(data = Obesity,
#                  method = meth_mnar,
#                  predictorMatrix = pred_mnar,
#                  pmm = T,
#                  visit=c("Height","FamOb","Age","Weight","BMI"),
#                  printFlag = FALSE)
#
# mice(data = Obesity,
#                  method = meth_mnar,
#                  predictorMatrix = pred_mnar,
#                  pmm = T,
#                  visit=c("Height","FamOb","Age","Weight","BMI"),
#                  printFlag = TRUE)

# TODO: find out why we get this error
# Fix it
# Submit to github
# Remove stuff above
```

```
imp_mnar <- mice(data = Obesity,
                 method = meth_mnar,
                 predictorMatrix = pred_mnar,
                 pmm = T,
                 visit=c("Height","FamOb","Age","Weight","BMI"),
                 printFlag = FALSE)
```

We check the convergence of the imputation models under the MAR and MNAR scenarios. Here we only display only the results for the MNAR scenario (Figure 22), but in none of them we found convergence problems.

```
ggmice::plot_trace(imp_mnar)
```

```
Trace plot could not be produced for variable(s):
  Cluster, Time, Gender
i No convergence diagnostics found.
```

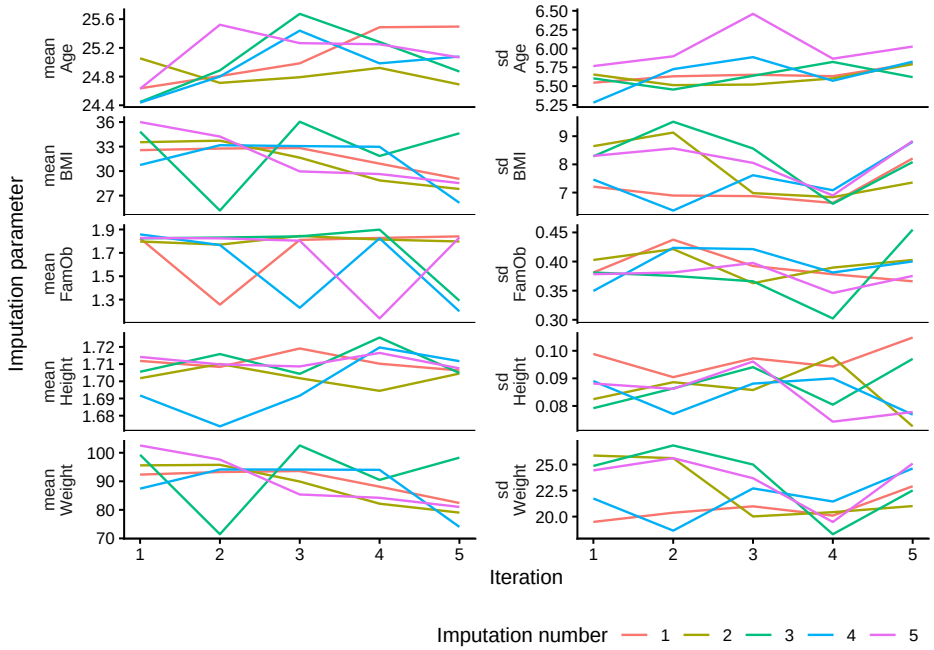
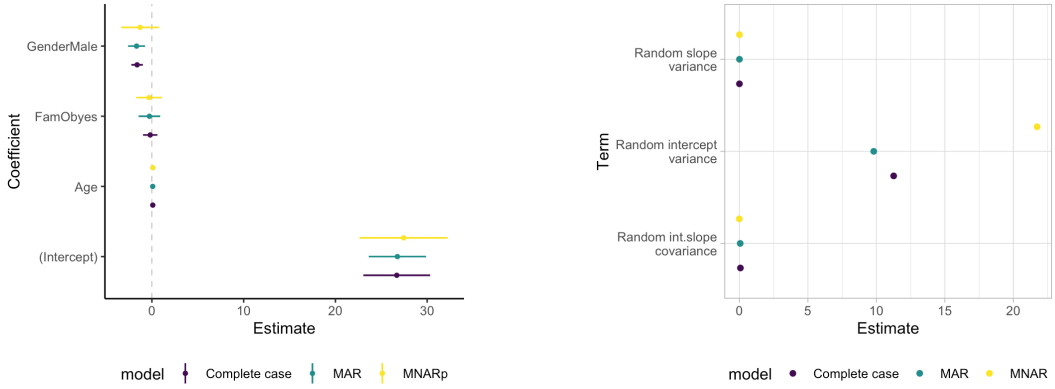


Figure 22: Trace plot for imputation under MNAR assumption

We proceed to compare the effects on the analysis model. We run the main analysis model on each of the imputed datasets as well as the dataset where the incomplete values are removed (Complete Case analysis, CC). Figure 23 shows the coefficient estimates along with the estimated variances of the random effects.

We observe minimal differences in the age effect, `FamObs`, across the various imputation models under consideration. However, we found that the effect of Gender becomes statistically non-significant under the MNAR assumption. By checking the intercept, we observe that under the missing not at random (MNAR) assumption, a higher average BMI is anticipated compared to under the missing at random (MAR) assumption. However, regarding the precision of the estimates, we notice that, in general, assuming MNAR leads to wider confidence intervals, and also larger variance of the random intercept compared to other scenarios.

5. Conclusion



(a) Estimated coefficients (b) Variances of the random effects

Figure 23: Estimates of the main model in the Obesity dataset

In this paper we explored the imputation process for incomplete multilevel datasets, with a primary focus on utilizing a multilevel model for analysis. We encourage users to consider insights provided by domain experts, to gain a better understanding of variable relationships and to chose the most appropriate imputation strategy to follow. This includes considerations of the (primary) research question and the corresponding data analysis, as the imputation strategy needs to be compatible with the data analysis. Employing clear data visualization is instrumental in comprehending the missing data patterns, investigating the likely missingness mechanism, and aiding in the selection of suitable imputation methods.

The **mice** and **mice**-based R packages offer a range of imputation methods tailored for multilevel data, easily adaptable to the dataset's structure. Before proceeding with the analysis of the imputed dataset, it is essential to assess the convergence of the imputation method. This evaluation can reveal issues such as circular dependencies, the need for additional iterations, or challenges associated with the chosen imputation method.

Disclaimer

The views expressed in this paper are the personal views of the authors and may not be understood or quoted as being made on behalf of or reflecting the position of the regulatory agency/agencies or organizations with which the authors are employed/affiliated.

Acknowledgments

This project has received funding from the European Union's Horizon 2020 research and innovation programme under ReCoDID grant agreement No 825746.

References

Andridge RR (2011). "Quantifying the impact of fixed effects modeling of clusters in multiple imputation for cluster randomized trials." *Biometrical Journal*, **53**(1), 57–74. ISSN 03233847. doi:10.1002/bimj.201000140. URL https://onlinelibrary.wiley.com/doi/10.1002/bimj.201000140.

Audigier V, Resche-Rigon MIE (2023). "Micemd: Multiple Imputation by Chained Equations with Multilevel Data."

Audigier V, White IR, Jolani S, Debray TPA, Quartagno M, Carpenter J, van Buuren S, Resche-Rigon M (2018). "Multiple Imputation for Multilevel Data with Continuous and Binary Variables." *Statistical Science*, **33**(2), 160–183. ISSN 0883-4237. 26770989.

Bailey BE, Andridge R, Shoben AB (2020). "Multiple Imputation by Predictive Mean Matching in Cluster-Randomized Trials." *BMC Medical Research Methodology*, **20**(1), 72. ISSN 1471-2288. doi:10.1186/s12874-020-00948-6.

Bartlett J, Keogh R, Bonneville EF (2022). "Multiple Imputation of Covariates by Substantive Model Compatible Fully Conditional Specification [R package smcfcs version 1.7.1]." URL https://CRAN.R-project.org/package=smcfcs.

Bates D, Maechler M, Bolker [aut B, cre, Walker S, Christensen RHB, Singmann H, Dai B, Scheipl F, Grothendieck G, Green P, Fox J, Bauer A, copyright on simulateformula) PNKs (2022). "Lme4: Linear Mixed-Effects Models Using 'Eigen' and S4."

Bolker B, Robinson D, Menne D, Gabry J, Buerkner P, Hua C, Petry W, Wiley J, Kennedy P, SE) ESB, Patil I, Arel-Bundock V (2021). "Broom.Mixed: Tidying Methods for Mixed Models."

Cai M, Van Buuren S, Vink G (2023). "Graphical and Numerical Diagnostic Tools to Assess Multiple Imputation Models by Posterior Predictive Checking." *Heliyon*, **9**(6), e17077. ISSN 24058440. doi:10.1016/j.heliyon.2023.e17077.

de Jong VMT, Moons KGM, Eijkemans MJC, Riley RD, Debray TPA (2021). "Developing More Generalizable Prediction Models from Pooled Studies and Large Clustered Data Sets." *Statistics in Medicine*, **40**(15), 3533–3559. ISSN 1097-0258. doi:10.1002/sim.8981.

Debray T, de Jong V (2021). "Metamisc: Meta-Analysis of Diagnosis and Prognosis Research Studies."

Drechsler J (2015). "Multiple Imputation of Multilevel Missing Data—Rigor Versus Simplicity." *Journal of Educational and Behavioral Statistics*, **40**(1), 69–95. ISSN 1076-9986. doi:10.3102/1076998614563393.

Eddings Wesley and Marchenko Y (2024). "How Can I Account for Clustering When Creating Imputations with Mi Impute?" https://www.stata.com/support/faqs/statistics/clustering-and-mi-impute/.

Enders CK, Mistler SA, Keller BT (2016). "Multilevel Multiple Imputation: A Review and Evaluation of Joint Modeling and Chained Equations Imputation." *Psychological Methods*, **21**(2), 222–240. ISSN 1939-1463. doi:10.1037/met0000063.

Galimard JE, Chevret S, Curis E, Resche-Rigon M (2018). "Heckman Imputation Models for Binary or Continuous MNAR Outcomes and MAR Predictors." *BMC Medical Research Methodology*, **18**(1), 90. ISSN 1471-2288. doi:10.1186/s12874-018-0547-1.

Gelman A, Hill J (2006). *Data Analysis Using Regression and Multilevel/Hierarchical Models*. Cambridge University Press. ISBN 978-1-139-46093-4.

Graham JW (2009). "Missing Data Analysis: Making It Work in the Real World." *Annual Review of Psychology*, **60**(1), 549–576. ISSN 0066-4308, 1545-2085. doi:10.1146/annurev.psych.58.110405.085530.

Grund S, Lüdtke O, Robitzsch A (2018). "Multiple Imputation of Missing Data for Multilevel Models: Simulations and Recommendations." *Organizational Research Methods*, **21**(1), 111–149. ISSN 1094-4281. doi:10.1177/1094428117703686.

Grund S, Lüdtke O, Robitzsch A (2021). "Multiple imputation of missing data in multilevel models with the R package mdmb: a flexible sequential modeling approach." *Behavior Research Methods*, **53**(6), 2631–2649. ISSN 1554-3528. doi:10.3758/s13428-020-01530-0. URL https://link.springer.com/10.3758/s13428-020-01530-0.

Grund S, Robitzsch A, Luedtke O (2023). "mitml: Tools for Multiple Imputation in Multilevel Modeling." URL https://cran.r-project.org/web/packages/mitml/index.html.

Hammon A (2022). "Multiple Imputation of Ordinal Missing Not at Random Data." *AStA Advances in Statistical Analysis*. ISSN 1863-818X. doi:10.1007/s10182-022-00461-9.

Hammon A, Zinn S (2020). "Multiple Imputation of Binary Multilevel Missing Not at Random Data." *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, **69**(3), 547–564. ISSN 0035-9254, 1467-9876. doi:10.1111/rssc.12401.

Hardt J, Herke M, Leonhart R (2012). "Auxiliary Variables in Multiple Imputation in Regression with Missing X: A Warning against Including Too Many in Small Sample Research." *BMC Medical Research Methodology*, **12**(1), 184. ISSN 1471-2288. doi:10.1186/1471-2288-12-184.

Hox J, van Buuren S, Jolani S (2015). "Incomplete Multilevel Data: Problems and Solutions." In J Harring, L Staplecton, S Beretvas (eds.), *Advances in Multilevel Modeling for Educational Research: Addressing Practical Issues Found in Real-World Applications*, CILVR Series on Latent Variable Methodology, pp. 39–62. Information Age Publishing Inc., Charlotte, NC. ISBN 978-1-68123-328-4.

Hox JJ, Moerbeek M, van de Schoot R (2017). *Multilevel Analysis: Techniques and Applications, Third Edition*. Routledge. ISBN 978-1-317-30868-3.

Jolani S (2018). "Hierarchical Imputation of Systematically and Sporadically Missing Data: An Approximate Bayesian Approach Using Chained Equations." *Biometrical Journal. Biometrische Zeitschrift*, **60**(2), 333–351. ISSN 1521-4036. doi:10.1002/bimj.201600220.

Localio AR, Berlin JA, Ten Have TR, Kimmel SE (2001). "Adjustments for Center in Multicenter Studies: An Overview." *Annals of Internal Medicine*, **135**(2), 112–123. ISSN 0003-4819. doi:10.7326/0003-4819-135-2-200107170-00012.

McNeish D, Kelley K (2019). "Fixed Effects Models versus Mixed Effects Models for Clustered Data: Reviewing the Approaches, Disentangling the Differences, and Making Recommendations." *Psychological Methods*, **24**(1), 20–35. ISSN 1939-1463, 1082-989X. doi:10.1037/met0000182.

Meng XL (1994). "Multiple-Imputation Inferences with Uncongenial Sources of Input." *Statistical Science*, **9**(4). doi:10.1214/ss/1177010269.

Mistler SA, Enders CK (2017). "A Comparison of Joint Model and Fully Conditional Specification Imputation for Multilevel Missing Data." *Journal of Educational and Behavioral Statistics*, **42**(4), 432–466. ISSN 1076-9986, 1935-1054. doi:10.3102/1076998617690869.

Molenberghs G, Kenward MG (2007). *Missing Data in Clinical Studies*. 1 edition. Wiley. ISBN 978-0-470-84981-1 978-0-470-51044-5. doi:10.1002/9780470510445.

Moons KG, Donders RA, Stijnen T, Harrell FE (2006). "Using the Outcome for Imputation of Missing Predictor Values Was Preferred." *Journal of Clinical Epidemiology*, **59**(10), 1092–1101. ISSN 08954356. doi:10.1016/j.jclinepi.2006.01.009.

Muñoz J, Efthimiou O, Audigier V, De Jong VMT, Debray TPA (2023a). "Multiple Imputation of Incomplete Multilevel Data Using Heckman Selection Models." *Statistics in Medicine*, p. sim.9965. ISSN 0277-6715, 1097-0258. doi:10.1002/sim.9965.

Muñoz J, Hufstedler H, Gustafson P, Bärnighausen T, De Jong VMT, Debray TPA (2023b). "Dealing with Missing Data Using the Heckman Selection Model: Methods Primer for Epidemiologists." *International Journal of Epidemiology*, **52**(1), 5–13. ISSN 0300-5771, 1464-3685. doi:10.1093/ije/dyac237.

Oberman H (2023). "ggmice: Visualizations for 'mice' with 'ggplot2'." URL https://cran.r-project.org/web/packages/ggmice/ggmice.pdf.

Pinheiro J, Bates D, DebRoy S, Sarkar D, Heisterkamp S (2023). "nlme: Linear and Nonlinear Mixed Effects Models." URL https://cran.r-project.org/web/packages/nlme/index.html.

Quartagno M, Grund S, Carpenter J (2019). "Jomo: A Flexible Package for Two-level Joint Modelling Multiple Imputation." *The R Journal*, **11**(2), 205–228. ISSN 2073-4859.

Rubin DB (1976). "Inference and Missing Data." *Biometrika*, **63**(3), 581–592. doi:10.2307/2335739.

Rubin DB (1987). *Multiple Imputation for Nonresponse in Surveys*. Wiley Series in Probability and Statistics, 1 edition. Wiley. ISBN 978-0-471-08705-2 978-0-470-31669-6. doi:10.1002/9780470316696. URL https://onlinelibrary.wiley.com/doi/book/10.1002/9780470316696.

Schafer JL, Yucel RM (2002). "Computational Strategies for Multivariate Linear Mixed-Effects Models With Missing Values." *Journal of Computational and Graphical Statistics*, **11**(2), 437–457. ISSN 1061-8600, 1537-2715. doi:10.1198/106186002760180608.

Speidel M, Drechsler J, Sakshaug JW (2018). "Biases in Multilevel Analyses Caused by Cluster-Specific Fixed-Effects Imputation." *Behavior Research Methods*, **50**(5), 1824–1840. ISSN 1554-3528. doi:10.3758/s13428-017-0951-1.

Steyerberg EW, Mushkudiani N, Perel P, Butcher I, Lu J, McHugh GS, Murray GD, Marmarou A, Roberts I, Habbema JDF, Maas AIR (2008). "Predicting Outcome after Traumatic Brain Injury: Development and International Validation of Prognostic Scores Based on Admission Characteristics." *PLOS Medicine*, **5**(8), e165. ISSN 1549-1676. doi:10.1371/journal.pmed.0050165.

Steyerberg EW, Nieboer D, Debray TP, van Houwelingen HC (2019). "Assessment of Heterogeneity in an Individual Participant Data Meta-Analysis of Prediction Models: An Overview and Illustration." *Statistics in Medicine*, **38**(22), 4290–4309. ISSN 1097-0258. doi:10.1002/sim.8296.

Taljaard M, Donner A, Klar N (2008). "Imputation Strategies for Missing Continuous Outcomes in Cluster Randomized Trials." *Biometrical Journal*, **50**(3), 329–345. ISSN 1521-4036. doi:10.1002/bimj.200710423.

Van Buuren S (2018). *Flexible Imputation of Missing Data*. Chapman and Hall/CRC.

van Buuren S, Groothuis-Oudshoorn K (2021). "Mice: Multivariate Imputation by Chained Equations."

White IR, Royston P (2009). "Imputing Missing Covariate Values for the Cox Model." *Statistics in Medicine*, **28**(15), 1982–1998. ISSN 0277-6715, 1097-0258. doi:10.1002/sim.3618.

Yucel RM (2008). "Multiple Imputation Inference for Multivariate Multilevel Continuous Data with Ignorable Non-Response." *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, **366**(1874), 2389–2403. doi:10.1098/rsta.2008.0038.

A2. Imputation strategies used in the analyses of the impact dataset

A2.1. Cluster indicator

```
meth <- mice::make.method(dati)
pred <- mice::make.predictorMatrix(dati)
pred.ci <- pred
pred.ci[,c("id","type")]<-0
plot_pred(pred.ci, method = meth, rotate = TRUE)

visit <- c("pupil","hypots","tsah","edh","hypox","ct")
impcind <- mice(dati, pred = pred.ci, meth =meth, visit=visit)
ggmice::plot_trace(impcind)
```

A2.2. Mltilevel imputation method

```
dati_h <- dati|> mutate(name = as.integer(name))
levels(dati_h$motor_score)
meth.h <- meth
meth.h[meth.h== "polyreg"] <- "2l.pmm"
meth.h[meth.h== "logreg"] <- "2l.bin"
pred.h <- pred
pred.h[pred.h == 1] <- 3 # we add cluster mean to level-1 variables
pred.h[,"id"]<-0
pred.h[,"name"] <- -2
pred.h[,"type"] <- 1 # we set level-2 type as a simple predictor (no mean)
plot_pred(pred.h, method = meth.h, rotate = TRUE)
visit <- c("pupil","hypots","tsah","edh","hypox","ct")
imphm <- mice(dati_h, pred = pred.h, meth = meth.h, visit = visit)
ggmice::plot_trace(himpimp)
# Backtransform the cluster variable
imphm <- complete(imphm, action='long', include=TRUE)
imphm$name <- as.factor(imphm$name)
levels(imphm$name)<-levels(dati$name)
imphm<-as.mids(imphm)
```

A2.3. Separate imputations

```
library(purrr)
pred.sep <- pred
pred.sep[,c("id","type","name")]<-0
plot_pred(pred.sep, method = meth, rotate = TRUE)
study_imp <- dati |>
  split(dati$name) |>
  purrr::map(~ mice(.x, pred=pred.sep,meth=meth,visit=visit))
names(study_imp) <- NULL
impsep<-study_imp[[1]]
for(i in 2:length(study_imp)){
  impsep <- mice::rbind(impsep,study_imp[[i]])
}

ggmice::plot_trace(impsep)
```

Appendix

A1. Two-level imputation methods

Table 1: *Some of the* `mice` *2l imputation methods*

Type	Package	Imputation method	Variable type	Description	Heteros-cedasticity	Systematic missing	Predictors type*
1-stage	mice	2l.norm	Continuous (level-1)	Uses LMM and requires at least 1 observation per cluster (Kasim, 1998).	Yes	No	0, 2
		2l.lmer		Uses an LMM estimated with lme4 (Jolani, 2017).	Yes	Yes	0, 1, 2
		2l.pan		LMM with homogeneous within group variances, relying on the pan R package (Schafer, 1997).	No	Yes	-1, 0, 1, 2, 3, 4
		2l.bin	Binary (level-1)	Better for small size clusters. It fits a GLMM with link (logit) using lme4 (Jolani, 2013).	Yes	Yes	0, 1, 2
		2l.only.(mean, norm, pmm)	Continuous, binary, categorical (level-2)	It can be used to check if level-2 variables are consistent across clusters.		Yes, sporadic only with 2l.mean	0, 1
	miceadds	2l.(continuous, binary, pmm)	Continuous, binary, categorical (level-1)	GMLM estimated with lme4 or the bayesian blme package (Snijders, 2012; Zinn, 2013).		Yes	0, 1, 2, 3, 4
	micemd	2l.glm.(norm, bin, pois)	Continuous, binary, count (level-1)	Bayesian GLMM with non-informative prior (Jeffrey). Better for few clusters and few observations per cluster. Binary logit link. (Jolani, 2017; Audigier 2018).	No	Yes	0, 2
		2l.jomo	Continuous, binary (level-1)	Bayesian MCMC approach, with a GMM based on conjugated priors (Quatagno Carpenter, 2016). Binary probit link. It relies on the jomo package.	Yes	Yes	0, 2
2-stage	Github repository	heckman.quart	Binary, ordered (level-1)	MNAR based on Heckman model using quadrature estimation with random intercept (Hammon 2020, 2022).	Yes	Yes	0, 1
	micemd	2l.2stage.(norm, bin, pmm, pois)	Continuous, binary, categorical, count (level-1)	Better for large size clusters. (Resche-Rignon, White, 2016; Audigier 2018). Binary logit link.	Yes	Yes	0, 2
		2l.2stage.heckman	Continuous, binary (level-1)	MNAR based on Heckman model estimated in two stages	Yes	Yes	-4, -3, 0, 2

LMM: Linear mixed model, GLMM: Generalized linear mixed model. * See Box 4 of the manuscript.

A2.4. Analysis on output of each strategy

```
# Save all datasets in a list
list_data <- list(dati|> filter(complete.cases(dati)), # complete case
                  impcind, # imputed with cluster indicator
                  imphm,# imputed with multilevel model
                  impsep) # imputation separated by cluster
# Run mixed model on each dataset
list_models <- lapply(list_data,
    FUN = function(x)
    with(x,lme4::glmer(mort ~ type + age + motor_score + pupil + ct
    + (1|name),
    family = binomial)))
# Plot models coefficients
plot_models(list_models,
            mod_name = c("Complete case",
        "Cluster indicator",
        "Hierarchical imputation",
        "Separate imputations"))
```

A3. Additional functions used in the analyses

A3.1. Visualize missing data (systematic vs. sporadic)

```
plot_na <- function() {
  dat <- expand.grid(rows = 1:7, cols = 1:6) |>
    cbind(
      text = c("1", "1", "2", "2", "3", "", "N", rep("", 35)),
      miss = c(rep("", 16), "NA", "NA", "", "", "", "NA", "", "", "NA", rep("", 17))
    )
  ggplot(dat, aes(x = cols, y = rows)) +
    geom_tile(fill = "white",
              color = "black",
              linewidth = 0.5) +
    geom_text(aes(label = text), color = "black", size = 3) +
    geom_text(
      aes(label = miss),
      color = mice:::mdc(2),
      family = "mono",
      fontface = "bold"
    ) +
    scale_x_continuous(
      breaks = 1:6,
      labels = c(
        "cluster",
        expression(X[1]),
        expression(X[2]),
        expression(X[3]),
        "...",
        expression(X[p])
      ),
      name = NULL,
      position = "top"
```

```
    ) +
    scale_y_continuous(
      breaks = 1:7,
      labels = c(1:5, "...", "n"),
      name = NULL,
      trans = "reverse"
    ) +
    theme_minimal() +
    theme(panel.grid = element_blank(),
          plot.margin = unit(c(0, 0, 0, 0), "pt"))
}
```

A3.2. Introduce more missingness to the popmis data

```
ampute.popmis <- function(dat){
  set.seed(1234)
  dat$prob <- 1 - plogis(-5 + 0.3 * (dat$sex==1) + 0.7 * dat$texp)
  dat$teachpop <- ifelse(rbinom(nrow(dat), 1, dat$prob),
                         NA, dat$teachpop) # MAR teach popularity
  dat$texp[sample(1:nrow(dat), 0.05*nrow(dat))] <- NA #MCAR texp
  dat$prob <- NULL
  dat$const <- NULL
  return(dat)
}
```

A3.3. Plot coefficients of an nlme model

```
plot_models <- function(list_models, mod_name){
  library(insight)
  library(tidyverse)
  library(broom.mixed)
  library(mitml)

  mod_table<-var_table<-NULL
  for(i in 1:length(list_models)){
    mod <- list_models[[i]]
    if(class(mod)[[1]]=="mira"){
      # For model estimated on imputed dataset ----
      ## Get coefficient estimates -----
      mod_tab <-
        summary(pool(mod),
                conf.int = TRUE)[,c("term","estimate","2.5 %","97.5 %","p.value")]
      ## Get variance estimates -----
      mitval <- data.frame(testEstimates(as.mitml.result(mod),
                                         extra.pars = TRUE)[["extra.pars"]])
      names_var <- rownames(mitval)

      vartable <-data.frame(Term=character(0),Value=numeric(0))
      if (length(names_var[grepl("^(Residual~~Residual)", names_var)])!=0){
        vartable <- vartable|>
          add_row(Term="Residual variance",
                  Value=estim[grepl("^(Residual~~Residual)", names_var)])
      }
```

```
  if (length(names_var[grepl("^(Intercept~~Intercept)",
                             names_var)])!=0){
    vartable <- vartable|>
      add_row(Term="Random intercept\nvariance",
              Value=estim[grepl("^(Intercept~~Intercept)",
                                names_var)]) }
  if (length(names_var[grepl("^(Intercept~~Slope)",
                             names_var)])!=0){
    vartable <- vartable|>
      add_row(Term=paste0("Random slope\nvariance ",
                          sub(".*\\|", "",
                              names_var[grepl("^(Intercept~~Intercept)",
                                              names_var)])),
              Value=estim[grepl("^(Intercept~~Slope)", names_var)])
  }

  if (length(names_var[grepl("^(Slope~~Slope)", names_var)])!=0)
  {
    vartable <- vartable|>
      add_row(Term=paste0("Random slope-intercept correlation ",

                          sub(".*\\|", "",
                              names_var[grepl("^(Slope~~Slope)", names_var)])),
              Value=estim[grepl("^(Slope~~Slope)",
                                names_var)])
  }

  if ((length(names_var[grepl("^(ICC)", names_var)])!=0)){
    vartable <- vartable|>
      add_row(Term="ICC",
              Value=estim[grepl("^(ICC)", names_var)])
  }

}else{

  # For model on CC -----
  ## Get coefficient estimates -----
  mod_tab<-tidy(mod,effects="fixed",conf.int=TRUE)
  mod_tab <-mod_tab[,c("term","estimate","conf.low",
                       "conf.high","p.value")]
  rownames(mod_tab)<-NULL
  ## Get variance estimates -----
  varcomp <- insight::get_variance(mod)
  vartable <- data.frame(Term="Residual variance",
                         Value=as.numeric(varcomp$var.residual))
  if (!is.null(varcomp$var.intercept)){
    vartable <- vartable|>
      add_row(Term="Random intercept\nvariance",
              Value=as.numeric(varcomp$var.intercept))
  }
  if (!is.null(varcomp$var.var.slope)){
    vartable <- vartable|>
      add_row(Term=paste0("Random slope var",
                          attr(varcomp$var.slope, "names")),
              Value=as.numeric(varcomp$var.intercept))
  }
```

```
    if (!is.null(varcomp$cor.slope_intercept)){
      vartable <- vartable|>
        add_row(Term=paste0("Random slope-intercept corr",
                            attr(varcomp$cor.slope_intercept, "names")),
                Value=as.numeric(varcomp$cor.slope_intercept))
    }
    if (!is.null(varcomp$var.intercept)){
      vartable <- vartable|>
        add_row(Term="ICC",
                Value=as.numeric(varcomp$var.intercept)/
                  (as.numeric(varcomp$var.intercept)+
                     as.numeric(varcomp$var.residual)))
    }
  }
  colnames(mod_tab)<-c("Coefficient","Estimate","lower","upper",
                       "p.value")
  mod_tab$model<-mod_name[i]
  mod_table<-rbind(mod_table,mod_tab)
  vartable$model <- mod_name[i]
  var_table <- rbind(var_table,vartable)
}

plot_coef <- ggplot(mod_table,
                    aes(x = Coefficient, y = Estimate,
                        ymin = lower, ymax = upper, group=model, color=model)) +
  geom_hline(yintercept=0,linetype="dashed",color="black",size=0.1)+
  geom_linerange( position = position_dodge(width = 0.8)) +
  geom_pointrange(size=0.1, position = position_dodge(width = 0.8))+
  coord_flip()+
  theme_classic()+theme(legend.position = "bottom")
+scale_color_viridis_d()
if(randisplay==T){
  var_table<-var_table|>filter(str_detect(Term, "^Random")) }
plot_var <- ggplot(var_table, aes(x=Term, y=Value, fill=model))+
  geom_bar(stat="identity", position="dodge")+
  coord_flip()+scale_fill_viridis_d()+
  xlab("Term")+ylab("Estimate")+
  theme_light()+theme(legend.position = "bottom")


return(list(plot_coef,plot_var))
}
```

Affiliation:

Hanne I. Oberman[1]
Methodology and Statistics
Padualaan 14
Utrecht The Netherlands
E-mail: h.i.oberman@uu.nl
URL: https://www.hanneoberman.github.io

Johanna Muñoz[1]
Julius Center for Health Sciences and Primary Care
Universiteitsweg 100
Utrecht The Netherlands

Gerko Vink
Methodology and Statistics
Padualaan 14
Utrecht The Netherlands

Thomas P.A. Debray
Utrecht The Netherlands

Valentijn M.T. de Jong
Julius Center for Health Sciences and Primary Care
Utrecht The Netherlands
E-mail: V.M.T.deJong-2@umcutrecht.nl

Journal of Statistical Software
published by the Foundation for Open Access Statistics
MMMMMM YYYY, Volume VV, Issue II
doi:10.18637/jss.v000.i00

http://www.jstatsoft.org/
http://www.foastat.org/
Submitted: yyyy-mm-dd
Accepted: yyyy-mm-dd